SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

CHENNAI-602105

COURSE CODE: ECA04

COURSE NAME: ANALOG CIRCUITS

MASTER RECORD

LIST OF EXPERIMENTS

LIST OF EXPERIMENTS	DURATION (EXPT CONDUCTION + EVALUATION)	CO	BL	WK	TYPE OF EXPT
1.Design of Voltage Series Feedback Amplifiers using BJT 1.1 Verification of Closed Loop Voltage Gain 1.2 Verification of Open Loop Voltage Gain 1.3 Frequency Response Analysis	180 Minutes	CO1	BL4	WK4	HARDWARE
2.Design of Voltage Shunt Feedback Amplifiers using BJT 2.1 Verification of Closed Loop Voltage Gain 2.2 Verification of Open Loop Voltage Gain 2.3 Frequency Response Analysis	180 Minutes	CO1	BL4	WK4	HARDWARE
3.Design of RC Phase Shift Oscillator and LC (Colpitts) Oscillator Circuit using BJT 3.1 Analysis of RC Phase Shift Oscillator 3.2 Analysis of LC Oscillator (Colpitts) 3.3 Percentage frequency error analysis of RC phase shift and LC oscillator	180 Minutes	CO2	BL4	WK4	HARDWARE
4.. Design and Analysis of Common Emitter Amplifiers using BJT 4.1 Verification of Voltage Gain 4.2 Determination of Lower and Upper Cutoff Frequencies 4.3 Frequency Response and Bandwidth Analysis	180 Minutes	CO3	BL4	WK4	HARDWARE
5.Frequency Response of CB Amplifier using LT Spice Simulator 5.1 Analysis of Voltage Gain 5.2 Input and Output Characteristic Analysis 5.3 Frequency Response Analysis	90 Minutes	CO3	BL4	WK4	SIMULATION

6.Design of Astable Multivibrator using BJT 6.1 Switching Analysis Astable Multivibrator 6.2 Measurement of Time Period and Frequency 6.3 Duty Cycle Analysis	180 Minutes	CO4	BL4	WK4	HAR DW ARE
7. Design and Simulation of Schmitt Trigger using LT Spice Simulator 7.1 Analyze function of Schmitt trigger and Determine bandwidth 7.2 Determination Of UTP and LTP Frequency Response Analysis 7.3 Analysis of Hysteresis Characteristics	90 Minutes	CO4	BL4	WK4	SIM ULA TIO N
8. Design and Simulation of UJT Relaxation Oscillator using LT Spice simulator 8.1 Analysis of UJT Relaxation Oscillator. 8.2 Determine Bandwidth 8.3 Determine Duty Cycle	90 Minutes	CO4	BL4	WK4	SIM ULA TIO N
9. Design of Hartley Oscillator using PCB Software 9.1 Design and Create PCB layout for Hartley Oscillator circuit. 9.2 Route PCB tracks with Manufacturing board 9.3 Generate fabrication files.	90 Minutes	CO5	BL6	WK6	SIM ULA TIO N
10.Design of Monostable Multivibrator using PCB Software 10.1.Design and Create PCB layout Monostable Multivibrator circuit. 10.2.Route PCB tracks with Manufacturing board 10.3. Generate fabrication files.	90 Minutes	CO5	BL6	WK6	SIM ULA TIO N

DESIGN AND ANALYSIS OF VOLTAGE SERIES FEEDBACK AMPLIFIER USING BJT

OBJECTIVES:

The experiment is divided into three engineering laboratory model tests:

1 Verification of Closed Loop Voltage Gain with feedback
2 Verification of Closed Loop Voltage Gain without feedback
3 Frequency Response Analysis for with feedback and without feedback amplifier

EXPERIMENT – 1.1

VERIFICATION OF CLOSED-LOOP VOLTAGE GAIN (WITH FEEDBACK) AMPLIFIER

Expected Learning Outcomes

After completion of this experiment, students will be able to:

1. Determine the closed-loop voltage gain experimentally.
2. Calculate feedback factor and loop gain.
3. Calculate Midband Frequency
4. Analyze the obtained values based on theoretical calculations and practical observations.

AIM:

To design and verify the closed-loop voltage gain of a voltage series feedback amplifier using BJT.

THEORETICAL BACKGROUND

A voltage series feedback amplifier is a negative feedback amplifier in which a fraction of the output voltage is fed back in series with the input signal. The feedback voltage opposes the input signal and stabilizes the amplifier gain.

DESIGN FORMULA:

For a voltage series feedback amplifier,

Closed-Loop Gain

$$A_f = \frac{A}{1 + A\beta}$$

Feedback Factor

$$\beta = \frac{A - A_f}{A \times A_f}$$

Loop Gain

$$A\beta = \frac{A}{A_f} - 1$$

Percentage Reduction in Gain

$$\%\ \text{Reduction in Gain} = \frac{A - A_f}{A} \times 100$$

Input Resistance with Feedback

$$R_{in,f} = R_{in}(1 + A\beta)$$

Output Resistance with Feedback

$$R_{out,f} = \frac{R_{out}}{1 + A\beta}$$

Bandwidth

$$BW = f_H - f_L$$

Gain-Bandwidth Product

$$A \times BW = A_f \times BW_f$$

Open-Loop Voltage Gain

$$A = \frac{V_o}{V_i}$$

Closed-Loop Voltage Gain (Experimental)

$$A_f = \frac{V_{of}}{V_i}$$

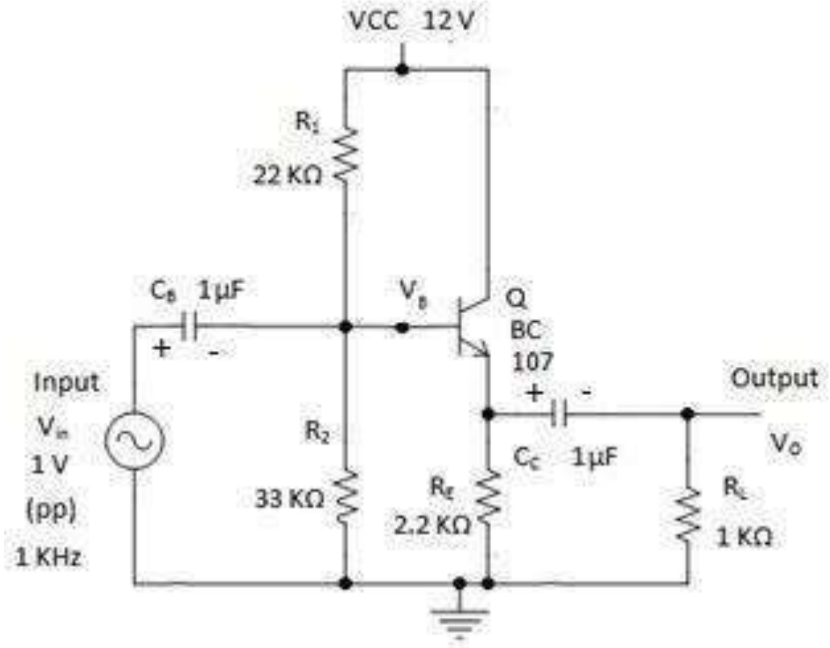
where:

- A = Open-loop gain
- A_f = Closed-loop gain
- β = Feedback factor
- R_{in} = Input resistance without feedback
- $R_{in,f}$ = Input resistance with feedback
- R_{out} = Output resistance without feedback
- $R_{out,f}$ = Output resistance with feedback
- f_L = Lower cutoff frequency
- f_H = Upper cutoff frequency
- BW = Bandwidth
- V_i = Input voltage
- V_o = Output voltage without feedback
- V_{of} = Output voltage with feedback

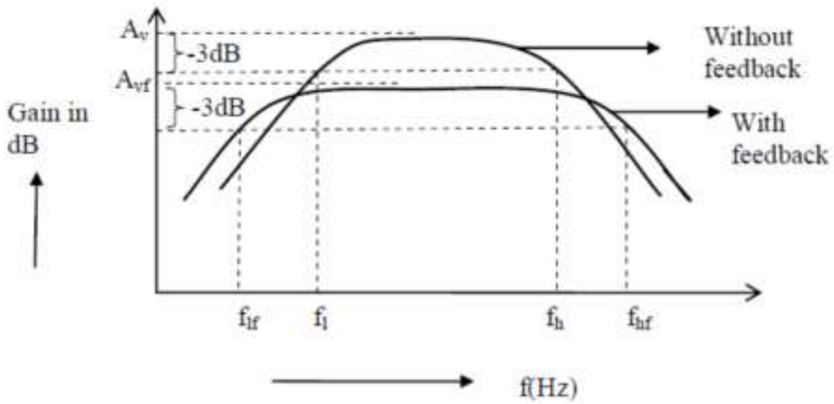
EXPERIMENTAL PROCEDURE:

1. Verify all components using a multimeter.
2. Assemble the voltage series feedback amplifier circuit on a breadboard.
3. Apply a DC supply voltage of 10 V.
4. Initially disconnect the feedback resistor.
5. Apply a sinusoidal input signal of 20 mV.
6. Measure input voltage and output voltage.
7. Calculate open-loop gain.
8. Connect the feedback network.
9. Measure input voltage and output voltage again.
10. Calculate closed-loop gain.
11. Determine feedback factor.
12. Compare theoretical and practical values.
13. Record all observations.

CIRCUIT DIAGRAM



MODEL GRAPH :



Bandwidth= f_l-f_h

TABULATION 1

Measurement of frequency response of voltage series with feedback amplifiers .

Vin=

Frequency (in Hz)	V_0 (Volts)	Gain= V_0/V_{in}	Gain (dB) = 20 log(V_o/V_{in})

RESULT

The closed-loop voltage gain of the voltage series with feedback amplifier was verified successfully.

EXPERIMENT – 1.2

VERIFICATION OF OPEN-LOOP VOLTAGE GAIN (WITHOUT FEEDBACK) AMPLIFIER

Expected Learning Outcomes

After completion of this experiment, students will be able to:

1. Determine the open-loop voltage gain experimentally.
2. Calculate Midband Frequency and voltage gain
3. Analyze the obtained values based on theoretical calculations and practical observations.

AIM:

To design and verify the open-loop voltage gain of a voltage series without feedback amplifier using BJT.

THEORETICAL BACKGROUND

A **BJT amplifier without feedback**, also known as an **open-loop amplifier**, amplifies the input signal without feeding any portion of the output back to the input. The voltage gain of the amplifier depends entirely on the transistor characteristics and the external circuit components such as resistors and the supply voltage. In an open-loop amplifier, the input signal is applied directly to the transistor, and the amplified output is obtained from the collector. Since there is no feedback network, the amplifier provides a high voltage gain. However, the gain is sensitive to changes in transistor parameters, temperature, and power supply variations, resulting in poor gain stability.

Expected Learning Outcomes

After completing this experiment, students will be able to:

1. Design and construct a BJT amplifier without feedback (open-loop amplifier).
2. Measure the input and output voltages of the amplifier.
3. Determine the open-loop voltage gain experimentally.
4. Analyze the frequency response of the amplifier.
5. Compare the theoretical and practical values of the voltage gain.
6. Understand the limitations of an amplifier without feedback, such as poor gain stability, higher distortion, and limited bandwidth.
7. Interpret the experimental results and relate them to the characteristics of an open-loop amplifier.

Formula:

For a **BJT amplifier without feedback (Open-Loop Amplifier)**, the commonly used formulas are:

1. Voltage Gain (Open-Loop)

$$\boxed{A_v = \frac{V_o}{V_{in}}}$$

Where:

- A_v = Voltage gain
- V_o = Output voltage (V)
- V_{in} = Input voltage (V)

2. Voltage Gain in Decibels (dB)

$$\boxed{Gain\,(dB) = 20\log_{10}\left(\frac{V_o}{V_{in}}\right)}$$

3. Input Resistance

$$\boxed{R_{in} = \frac{V_{in}}{I_{in}}}$$

Where:

- R_{in} = Input resistance (Ω)
- I_{in} = Input current (A)

4. Output Resistance

$$\boxed{R_{out} = \frac{V_{out}}{I_{out}}}$$

Where:

- R_{out} = Output resistance (Ω)
- I_{out} = Output current (A)

5. Bandwidth

$$\boxed{BW = f_H - f_L}$$

Where:

- f_H = Upper cutoff frequency (Hz)
- f_L = Lower cutoff frequency (Hz)

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6. Current Gain (if required)

$$\boxed{A_i = \frac{I_o}{I_{in}}}$$

7. Power Gain (if required)

$$\boxed{A_p = A_v \times A_i}$$

or

$$\boxed{A_p = \frac{P_o}{P_{in}}}$$

EXPERIMENTAL PROCEDURE

1. Verify all the components and the BJT transistor using a multimeter.
2. Assemble the BJT amplifier circuit on a breadboard as per the circuit diagram, ensuring that the feedback resistor is disconnected (without feedback configuration).
3. Connect the DC power supply and set the supply voltage to **10 V**.
4. Connect the function generator to the amplifier input and apply a sinusoidal input signal of **20 mV** at **1 kHz**.
5. Measure the input voltage ((V_{in})) using the CRO.
6. Measure the output voltage ((V_{o})) using the CRO.
7. Record the output voltage at each frequency.

TABULATION:

Measurement of frequency response of voltage series without feedback amplifiers .

Vin=

Frequency (in Hz)	**V_0 (Volts)**	**Gain= V_0/V_{in}**	**Gain (dB) = 20 log(V_o/V_{in})**

RESULT

The open-loop voltage gain of the voltage series without feedback amplifier was verified successfully.

EXPERIMENT – 1.3

FREQUENCY RESPONSE ANALYSIS

Expected Learning Outcomes

After completion of this experiment, students will be able to:

1. Determine lower and upper cutoff frequencies.
2. Evaluate bandwidth improvement due to feedback.

Aim

To obtain the frequency response of a voltage series feedback amplifier and determine its bandwidth.

THEORY

Negative feedback improves amplifier bandwidth while reducing gain.

The gain-bandwidth product remains approximately constant.

Gain:

$$A_v = \frac{V_o}{V_i}$$

Gain in dB:

$$Gain(dB) = 20\log_{10}\left(\frac{V_o}{V_i}\right)$$

Bandwidth:

$$BW = f_H - f_L$$

Where:

fH = Upper cutoff frequency

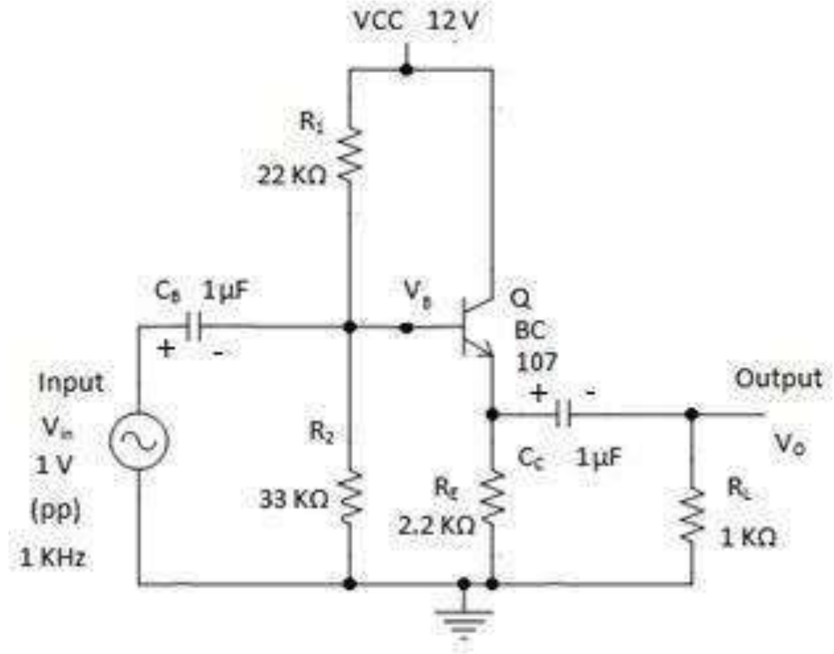
fL = Lower cutoff frequency

DETAILED PROCEDURE

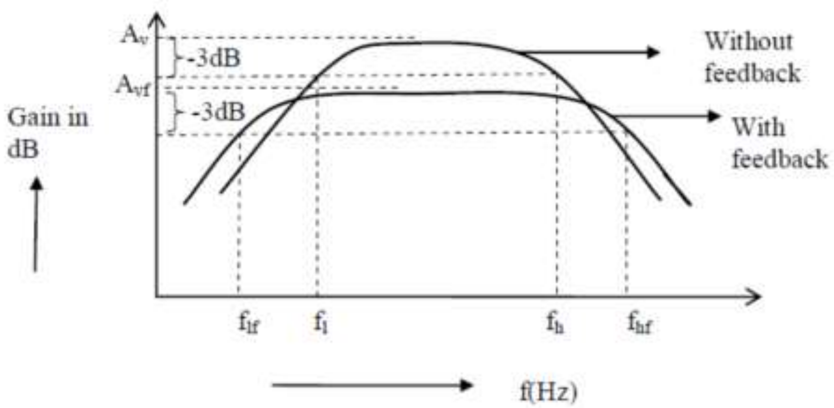
1. Assemble the voltage series feedback amplifier.
2. Apply a constant input signal of 20 mV.
3. Set frequency to 100 Hz.
4. Measure output voltage.
5. Increase frequency gradually.

6. Record output voltage for each frequency.
7. Calculate gain.
8. Convert gain into dB.
9. Plot frequency response curve.
10. Determine cutoff frequencies.
11. Calculate bandwidth.

CIRCUIT DIAGRAM



MODEL GRAPH :



Bandwidth= f_l-f_h

VERIFICATION TABLE

Parameter	CLOSED LOOP VOLTAGE	OPEN LOOP VOLTAGE
Lower Cutoff Frequency		
Upper Cutoff Frequency		
Band width		

RESULT

The frequency response characteristics of the voltage series feedback amplifier with feedback and without feedback (closed loop and open loop) were plotted successfully.

DESIGN AND ANALYSIS OF VOLTAGE SHUNT FEEDBACK AMPLIFIER USING BJT

OBJECTIVES:

The experiment is divided into three engineering laboratory model tests:

1.Verification of Closed Loop Voltage Gain

2. Verification of Open Loop Voltage Gain

3.Frequency Response Analysis

EXPERIMENT – 2.1

CLOSED-LOOP VOLTAGE GAIN (WITH FEEDBACK) AMPLIFIER

Expected Learning Outcomes

After completion of this experiment, students will be able to:

1. Design a voltage shunt feedback amplifier using BJT.
2. Measure open-loop and closed-loop gains.
3. Determine feedback factor and loop gain.
4. Analyze the effect of shunt mixing on amplifier performance.
5. Compare theoretical and practical results.

AIM:

To design and verify the closed-loop voltage gain of a voltage shunt feedback amplifier using BJT.

THEORETICAL BACKGROUND

A voltage shunt feedback amplifier samples the output voltage and feeds a fraction of it back in parallel (shunt) with the input signal.

Because the feedback signal is connected in parallel with the input:

- Gain decreases
- Input resistance decreases
- Output resistance decreases
- Bandwidth increases
- Linearity improves

The closed-loop gain is given by

DESIGN FORMULA:

For a voltage shuntfeedback amplifier,

$$A_f = \frac{A}{1 + A\beta}$$

Where:

A = Open-loop gain

A_f = Closed-loop gain

β = Feedback factor

$A\beta$ = Loop gain

Closed Loop Gain

$$A_f = \frac{A}{1 + A\beta}$$

Feedback Factor

$$\beta = \frac{V_f}{V_o}$$

Loop Gain

$$A\beta = A \times \beta$$

Midband Gain (Closed-Loop)

$$A_{f(mid)} = \frac{V_{of}}{V_i}$$

(i) Closed-Loop Gain

$$A_f = \frac{A}{1 + A\beta}$$

(ii) Input Resistance with Feedback

For Voltage Shunt Feedback:

$$R_{in,f} = \frac{R_{in}}{1 + A\beta}$$

(iii) Output Resistance with Feedback

For Voltage Shunt Feedback:

$$R_{out,f} = \frac{R_{out}}{1 + A\beta}$$

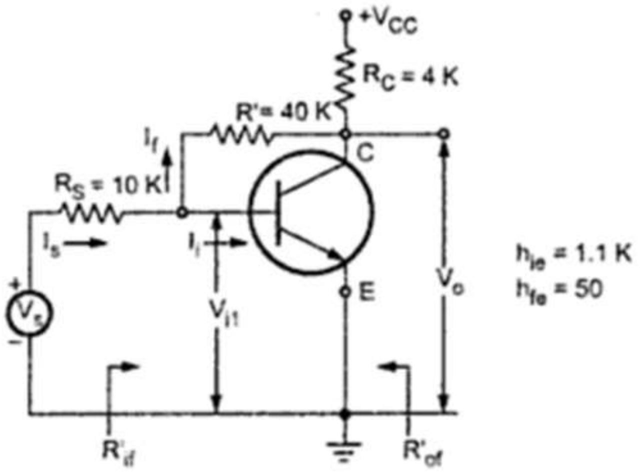
(iv) Percentage Reduction in Gain

$$\% \text{ Reduction in Gain} = \frac{A - A_f}{A} \times 100$$

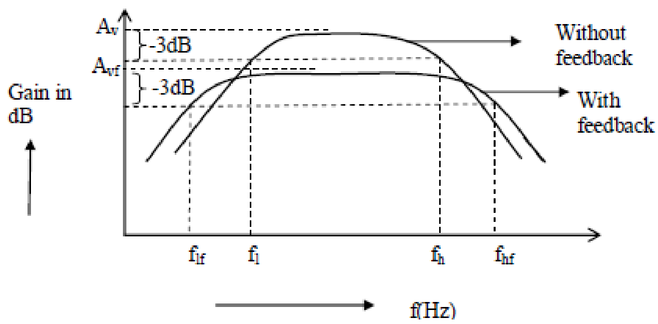
EXPERIMENTAL PROCEDURE:

1. EXPERIMENTAL PROCEDURE
2. Verify all components using a multimeter.
3. Assemble the voltage shunt feedback amplifier circuit.
4. Apply a DC supply voltage of 10 V.
5. Initially disconnect the feedback resistor.
6. Apply a sinusoidal input signal of 20 mV.
7. Measure input and output voltages.
8. Calculate open-loop gain.
9. Connect the feedback network.
10. Measure output voltage again.
11. Calculate closed-loop gain.
12. Determine feedback factor.
13. Compare theoretical and practical values.
14. Record observations.

CIRCUIT DIAGRAM



MODEL GRAPH :



Bandwidth= f_l-f_h

TABULATION

Measurement of frequency response of voltage shunt feedback amplifiers .

Vin=

Frequency (in Hz)	V_0 (Volts)	Gain= V_0/V_{in}	Gain (dB) = 20 log(V_o/V_{in})

RESULT

The closed-loop voltage gain of the voltage shunt feedback amplifier was verified successfully.

EXPERIMENT – 2.2

VERIFICATION OF OPEN LOOP VOLTAGE GAIN

Expected Learning Outcomes

After completion of this experiment, students will be able to:

1. Understand the basic working principle of a **shunt amplifier** and its open-loop configuration
2. Identify the difference between **open-loop and closed-loop amplifier operation**
3. Measure input and output voltages using CRO/DSO and DMM accurately
4. Calculate **open-loop voltage gain** using experimental readings
5. Analyze how amplification behaves without feedback and recognize its limitations
6. Develop skills in setting up and troubleshooting analog amplifier circuits

7. Understand the importance of feedback in improving amplifier stability and performance
8. **8.** Compare theoretical and practical gain values and interpret variations

Aim

To verify the open-loop voltage gain of a shunt amplifier by measuring the input and output voltages.

THEORETICAL BACKGROUND

A **shunt amplifier** is an amplifier configuration in which the feedback is connected in parallel (shunt) with the input. Before applying feedback, the amplifier operates in **open-loop mode**, where the output depends only on the intrinsic gain of the amplifier. The **open-loop voltage gain** is the ratio of the output voltage to the input voltage without any feedback network connected. Measuring the open-loop gain helps understand the amplifier's performance before studying the effects of feedback. Open-loop gain is usually high, but it is less stable and more sensitive to component variations and noise. Negative feedback is later used to improve stability, bandwidth, and linearity.

Open-Loop Voltage Gain

$$A_v = \frac{V_{out}}{V_{in}}$$

where:

- A_v = Open-loop voltage gain
- V_{out} = Output voltage (V)
- V_{in} = Input voltage (V)

Gain in Decibels (Optional)

$$A_v(\mathrm{dB}) = 20\log_{10}\left(\frac{V_{out}}{V_{in}}\right)$$

EXPERIMENTAL PROCEDURE

1. Connect the circuit according to the open-loop shunt amplifier circuit diagram.
2. Ensure the feedback connection is removed.
3. Switch ON the regulated DC power supply.
4. Apply a sinusoidal input signal (typically **100 mV to 200 mV** at **1 kHz**) from the function generator.
5. Measure the input voltage (VinV_{in}Vin) using the CRO/DSO.
6. Measure the output voltage (VoutV_{out}Vout).
7. Record the readings in the observation table.
8. Calculate the open-loop voltage gain using the formula.
9. Compare the calculated value with the theoretical value.

TABULATION

Measurement of frequency response of voltage shunt feedback amplifiers .

Vin=

Frequency (in Hz)	**V_0 (Volts)**	**Gain= V_0/V_{in}**	**Gain (dB) = 20 log(V_o/V_{in})**

RESULT

The open-loop voltage gain of the shunt amplifier was determined by measuring the input and output voltages and verified using:

EXPERIMENT – 2.3

FREQUENCY RESPONSE ANALYSIS

Expected Learning Outcomes

After completion of this experiment, students will be able to:

1. Plot frequency response characteristics.
2. Calculate gain in decibels.
3. Determine amplifier bandwidth.
4. Compare bandwidth with and without feedback.

Aim

To obtain the frequency response of a voltage shunt feedback amplifier and determine its bandwidth.

THEORY

Negative feedback improves amplifier bandwidth while reducing gain.

The gain-bandwidth product remains approximately constant.

Gain:

$$A_v = \frac{V_o}{V_i}$$

Gain in dB:

$$Gain(dB) = 20\log_{10}\left(\frac{V_o}{V_i}\right)$$

Bandwidth:

$$BW = f_H - f_L$$

Where:

fH = Upper cutoff frequency

fL = Lower cutoff frequency

(i) Feedback Factor

From

$$A_f = \frac{A}{1 + A\beta}$$

Therefore,

$$\beta = \frac{A - A_f}{A \times A_f}$$

(ii) Loop Gain

$$A\beta = \frac{A}{A_f} - 1$$

(iii) New Input Resistance

$$R_{in,f} = \frac{R_{in}}{1 + A\beta}$$

(iv) Feedback Resistor Network

For a resistive voltage divider:

$$\beta = \frac{R_1}{R_1 + R_f}$$

or

$$R_f = R_1 \left(\frac{1 - \beta}{\beta} \right)$$

where:

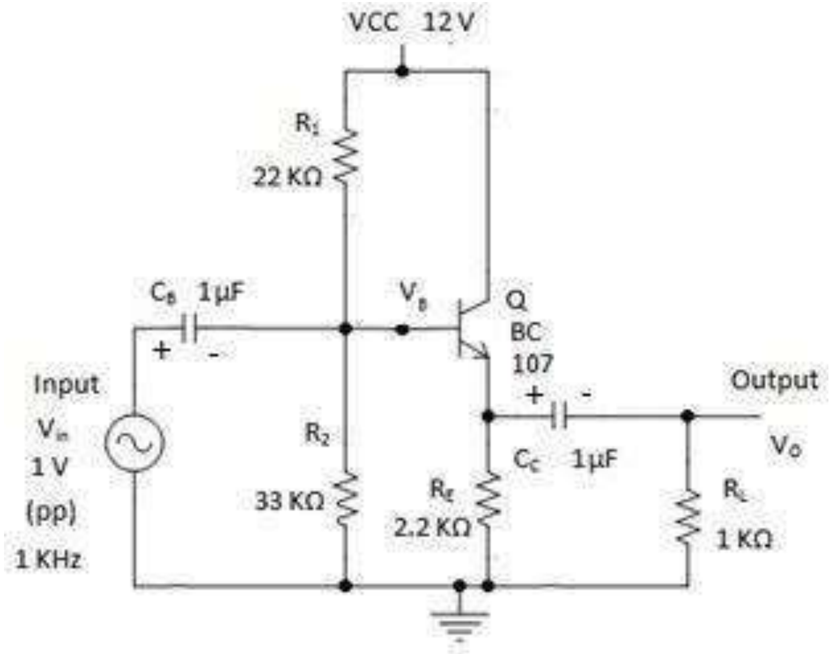
- R_f = Feedback resistor
- R_1 = Input-side resistor of feedback network

A	Open-Loop Voltage Gain (Gain without feedback)
A_f	Closed-Loop Voltage Gain (Gain with feedback)
β	Feedback Factor
$A\beta$	Loop Gain (Return Difference)
R_{in}	Input Resistance without Feedback
$R_{in,f}$	Input Resistance with Feedback
R_{out}	Output Resistance without Feedback
$R_{out,f}$	Output Resistance with Feedback
R_{of}	Effective Output Resistance with Feedback
R_L	Load Resistance
R_f	Feedback Resistor
R_1	Input-side Resistor in Feedback Network
D	Distortion without Feedback
D_f	Distortion with Feedback
V_i	Input Voltage
V_o	Output Voltage without Feedback
V_{of}	Output Voltage with Feedback
I_i	Input Current
I_o	Output Current

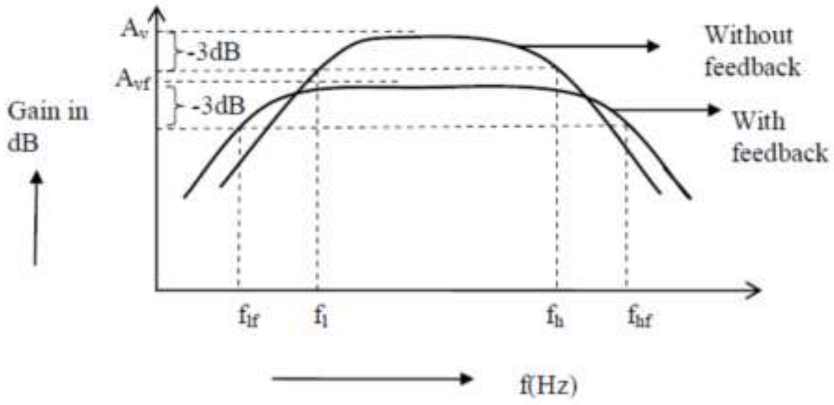
DETAILED PROCEDURE

1. Connect the circuit without feedback.
2. Apply a constant input signal of 20 mV.
3. Measure output voltage at various frequencies.
4. Calculate gain and gain in dB.
5. Connect feedback resistor RF.
6. Repeat measurements.
7. Plot frequency response curves.
8. Determine cutoff frequencies.
9. Calculate bandwidth.

CIRCUIT DIAGRAM



MODEL GRAPH :



Bandwidth= f_l-f_h

TABULATION

Parameter	CLOSED LOOP VOLTAGE	OPEN LOOP VOLTAGE
Lower Cutoff Frequency		
Upper Cutoff Frequency		
Band width		

RESULT

The frequency response characteristics of the voltage shunt feedback amplifier were plotted successfully and the bandwidth improvement due to feedback was verified.

EXPERIMENT – 3

DESIGN AND ANALYSIS OF RC PHASE SHIFT OSCILLATOR AND LC OSCILLATOR USING BJT

OBJECTIVES

The experiment is divided into three engineering laboratory model tests:

1.Analysis of RC Phase Shift Oscillator

2. Analysis of LC Oscillator (Colpitts)

3. Percentage frequency error analysis of RC phase shift and LC oscillator

EXPERIMENT – 3.1

ANALYSIS OF RC PHASE SHIFT OSCILLATOR

Expected Learning Outcomes

After completion of this experiment, students will be able to:

1. Design an RC Phase Shift Oscillator using BJT.
2. Understand Barkhausen Criterion.
3. Measure oscillation frequency.
4. Analyze feedback networks in oscillators.
5. Compare theoretical and practical oscillation frequencies.

Aim

To design and verify the operation of an RC Phase Shift Oscillator using BJT.

THEORETICAL BACKGROUND

An oscillator is an electronic circuit that generates periodic waveforms without requiring an external input signal.

An RC Phase Shift Oscillator consists of:

1. A transistor amplifier stage.
2. Three RC sections providing phase shift.
3. Positive feedback network.

The transistor amplifier provides:180°phase shift.
The RC network provides:180°additional phase shift.

Hence total phase shift becomes:360∘ which satisfies Barkhausen Criterion.

DESIGN FORMULA

Frequency of Oscillation

$$f = \frac{1}{2\pi RC\sqrt{6}}$$

where

R = Resistance of each RC section

C = Capacitance of each RC section

2. Resistance Calculation

$$R = \frac{1}{2\pi f_o C\sqrt{6}}$$

---

3. Capacitance Calculation

$$C = \frac{1}{2\pi f_o R\sqrt{6}}$$

---

4. Condition for Sustained Oscillations (Barkhausen Criterion)

$$A\beta = 1$$

5. Feedback Factor of Three-Section RC Network

$$\beta = \frac{1}{29}$$

---

6. Minimum Amplifier Gain Required

$$A_{min} = 29$$

---

7. Gain in Decibels

$$G_{dB} = 20\log_{10}(A)$$

8. Gain Margin

$$\text{Gain Margin} = 20\log_{10}\left(\frac{A}{A_{min}}\right)$$

---

9. Transistor Current Gain

$$\beta = \frac{I_C}{I_B}$$

---

10. Base Current

$$I_B = \frac{I_C}{\beta}$$

---

11. Collector Current

$$I_C = \beta I_B$$

---

12. Collector Resistor

$$R_C = \frac{V_{CC} - V_{CE}}{I_C}$$

13. Emitter Resistor

$$R_E = \frac{V_E}{I_E}$$

14. Base Voltage

$$V_B = V_E + V_{BE}$$

15. Voltage Divider Bias Resistors

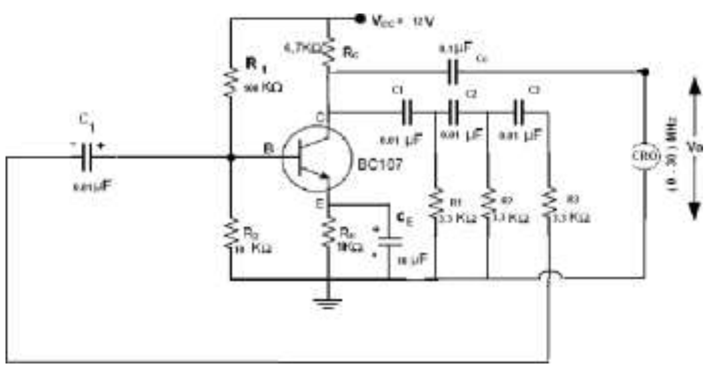
$$R_1 = \frac{V_{CC} - V_B}{I_{R1}}$$

$$R_2 = \frac{V_B}{I_{R2}}$$

Procedure:

1. Verify all components.
2. Assemble RC phase shift oscillator circuit.
3. Apply DC supply voltage.
4. Verify transistor biasing.
5. Observe output waveform on CRO.
6. Measure oscillation frequency.
7. Compare theoretical and practical values.
8. Record observations.

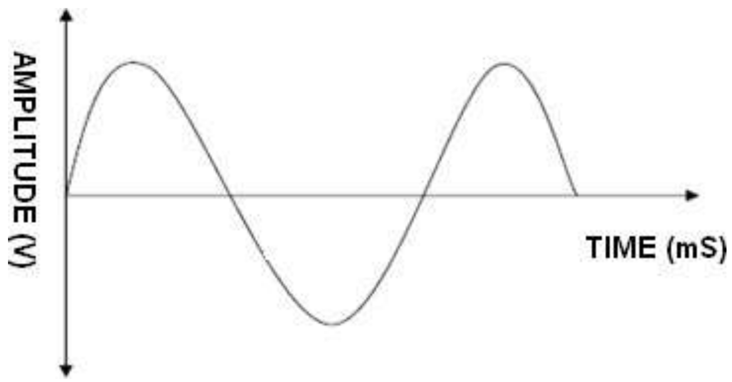
CIRCUIT DIAGRAM:



Tabulation:

Oscillator	AMPLITUDE(V)	TIME PERIOD (ms)	FREQUENCY(Hz)
RC THEORITICAL			
RC PRACTIAL			

Model Graph:



RESULT

The RC Phase Shift Oscillator was designed successfully and the oscillation frequency was verified.

EXPERIMENT – 3.2

ANALYSIS OF OF LC OSCILLATOR

Expected Learning Outcomes

After completion of this experiment, students will be able to:

1. Design an LC oscillator (Colpitts).
2. Measure resonant frequency.
3. Analyze energy transfer between L and C.
4. Verify oscillation conditions.
5. Compare theoretical and practical frequencies.

Aim

To design and verify the operation of an LC oscillator (Colpitts) using BJT.

THEORETICAL BACKGROUND

LC oscillators generate sinusoidal waveforms using an inductive-capacitive tank circuit. The oscillation frequency depends on: Inductance (L) and Capacitance (C)

DESIGN FORMULA:

Resonant Frequency

$$f = \frac{1}{2\pi\sqrt{LC}}$$

where

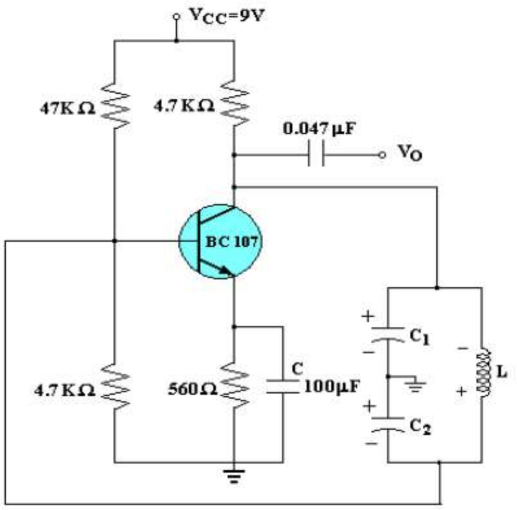
L = Inductance

C = Capacitance

EXPERIMENTAL PROCEDURE

1. Assemble LC oscillator circuit.
2. Verify component values.
3. Apply supply voltage.
4. Observe sinusoidal waveform.
5. Measure frequency.
6. Compare theoretical and practical frequencies.
7. Record observations.

Circuit diagram:

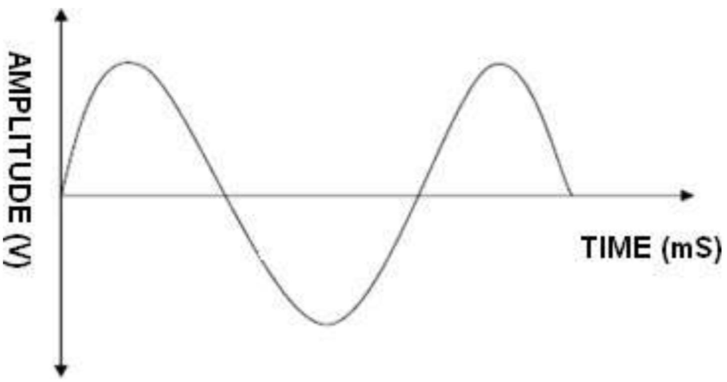


Colpitts Oscillator

Tabulation:

Oscillator	AMPLITUDE(V)	TIME PERIOD (ms)	FREQUENCY(Hz)
LC THEORITICAL			
LC PRACTIAL			

Model Graph:



RESULT

The LC oscillator was designed successfully and the oscillation frequency was verified.

EXPERIMENT – 3.3

PERCENTAGE FREQUENCY ERROR ANALYSIS OF RC PHASE SHIFT AND LC OSCILLATOR

Expected Learning Outcomes

After completion of this experiment, students will be able to:

1. Compare RC and LC oscillators.
2. Analyze frequency stability.
3. Evaluate oscillator performance in terms of Percentage Frequency Error.

Aim

To analyze the percentage frequency error of RC phase shift and LC oscillator

THEORY

- RC Oscillator
- Suitable for audio frequencies.
- Uses resistors and capacitors.
- Simple construction.
- LC Oscillator
- Suitable for radio frequencies.
- Uses inductors and capacitors.
- Better frequency stability.

DESIGN FORMULA:

RC Oscillator

$$f = \frac{1}{2\pi RC\sqrt{6}}$$

LC Oscillator

$$f = \frac{1}{2\pi\sqrt{LC}}$$

LC Colpitts Oscillator (BJT) – Required Formulae Only

1. Equivalent Capacitance of Colpitts Tank Circuit

$$C_T = \frac{C_1 C_2}{C_1 + C_2}$$

where:

- C_1 = Capacitor 1
- C_2 = Capacitor 2
- C_T = Equivalent capacitance

---

2. Frequency of Oscillation

$$f_o = \frac{1}{2\pi\sqrt{LC_T}}$$

where:

- L = Inductance
- C_T = Equivalent capacitance

---

3. Inductance Calculation

$$L = \frac{1}{(2\pi f_o)^2 C_T}$$

4. Equivalent Capacitance Calculation

$$C_T = \frac{1}{(2\pi f_o)^2 L}$$

---

5. Barkhausen Criterion

$$A\beta = 1$$

where:

- A = Amplifier Gain
- β = Feedback Factor

---

6. Feedback Factor

$$\beta = \frac{C_1}{C_2}$$

or

$$\beta = \frac{C_2}{C_1}$$

(depending on the feedback connection)

---

7. Minimum Gain Condition

$$A \geq \frac{C_2}{C_1}$$

---

8. Transistor Current Gain

$$\beta = \frac{I_C}{I_B}$$

---

9. Base Current

$$I_B = \frac{I_C}{\beta}$$

10. Collector Current

$$I_C = \beta I_B$$

---

11. Collector Resistor

$$R_C = \frac{V_{CC} - V_{CE}}{I_C}$$

---

12. Emitter Resistor

$$R_E = \frac{V_E}{I_E}$$

13. Base Voltage

$$V_B = V_E + V_{BE}$$

14. Voltage Divider Bias

$$R_1 = \frac{V_{CC} - V_B}{I_{R1}}$$

$$R_2 = \frac{V_B}{I_{R2}}$$

15. Output Voltage

$$V_{pp} = 2V_p$$

where:

- V_{pp} = Peak-to-Peak Voltage
- V_p = Peak Voltage

16. Frequency Stability

Percentage frequency change:

$$\%\Delta f = \frac{f_2 - f_1}{f_1} \times 100$$

17. Frequency Stability Factor

$$S_f = \frac{\Delta f}{f_o}$$

18.FREQUENCY OF ERROR:

$$\%\ \mathrm{Error} = \frac{|f_{\mathrm{theoretical}} - f_{\mathrm{measured}}|}{f_{\mathrm{theoretical}}} \times 100$$

Measure **Frequency Stability** by recording the frequency several times over a few minutes and calculating:

$$\Delta f = f_{\max} - f_{\min}$$

or

$$\% \text{ Stability} = \frac{\Delta f}{f_{\text{average}}} \times 100$$

IF IN QUESTION PAPER THEY MENTION LC OSCILLATOR USE THIS FORMULAS:

1. Frequency of Oscillation

$$f_o = \frac{1}{2\pi\sqrt{L_T C_T}}$$

where

$$L_T = L_1 + L_2 + 2M$$

For negligible mutual inductance,

$$L_T = L_1 + L_2$$

If two capacitors are connected in series:

$$C_T = \frac{C_1 C_2}{C_1 + C_2}$$

If

$$C_1 = C_2 = C$$

then

$$C_T = \frac{C}{2}$$

3. Inductance Calculation

$$L_T = \frac{1}{(2\pi f_o)^2 C_T}$$

---

4. Capacitance Calculation

$$C_T = \frac{1}{(2\pi f_o)^2 L_T}$$

---

5. Barkhausen Criterion

$$A\beta = 1$$

6. Feedback Ratio of Hartley Oscillator

$$\beta = \frac{L_1}{L_2}$$

or

$$\beta = \frac{L_2}{L_1}$$

(depending on the feedback winding arrangement)

---

7. Minimum Gain Condition

$$A \geq \frac{L_2}{L_1}$$

or

$$A \geq \frac{L_1}{L_2}$$

---

8. Transistor Current Gain

$$\beta = \frac{I_C}{I_B}$$

---

9. Base Current

$$I_B = \frac{I_C}{\beta}$$

10. Collector Current

$$I_C = \beta I_B$$

11. Collector Resistor

$$R_C = \frac{V_{CC} - V_{CE}}{I_C}$$

12. Emitter Resistor

$$R_E = \frac{V_E}{I_E}$$

13. Base Voltage

$$V_B = V_E + V_{BE}$$

14. Voltage Divider Bias

Upper Bias Resistor

$$R_1 = \frac{V_{CC} - V_B}{I_{R1}}$$

Lower Bias Resistor

$$R_2 = \frac{V_B}{I_{R2}}$$

15. Output Voltage

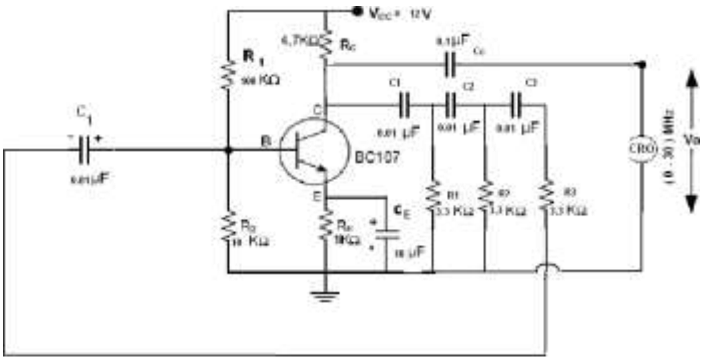
$$V_{pp} = 2V_p$$

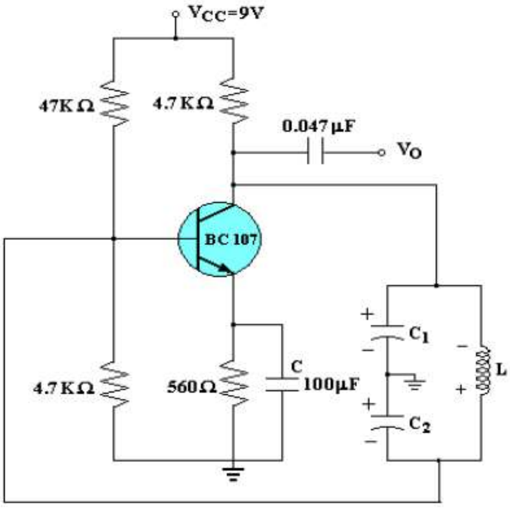
where

- V_{pp} = Peak-to-Peak Output Voltage
- V_p = Peak Output Voltage

EXPERIMENTAL PROCEDURE

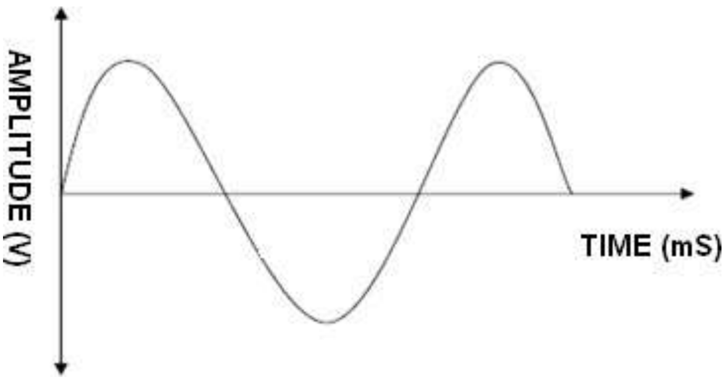
1. Assemble RC oscillator.
2. Measure oscillation frequency.
3. Assemble LC oscillator.
4. Measure oscillation frequency.
5. Compare results.
6. Analyze waveform quality.
7. Record observations.





Colpitts Oscillator

Model Graph:



Tabulation:

Oscillator	Time Period (µs	Frequency (kHz)	Amplitude (Vpp)	% Error/ Stability
RC PHSE SHIFT				
LC				

RESULT

The percentage frequency error of RC and LC oscillators were analyzed successfully. The LC oscillator exhibited superior frequency stability compared to the RC Phase Shift Oscillator.

EXPERIMENT – 4

FREQUENCY RESPONSE OF COMMON EMITTER AMPLIFIER USING BJT

OBJECTIVES

The experiment is divided into three engineering laboratory model tests:

1. Verification of Voltage Gain
2. Determination of Lower and Upper Cutoff Frequencies
3. Frequency Response and Bandwidth Analysis

EXPERIMENT – 4.1

ANALYSIS OF VOLTAGE GAIN FOR COMMON EMITTER AMPLIFIER

Expected Learning Outcomes

After completion of this experiment, students will be able to:

1. Design a CE amplifier.
2. Measure voltage gain experimentally.
3. Compare theoretical and practical gain values.
4. Understand transistor amplification principles.
5. Analyze the effect of transistor parameters on gain.

Aim

To design and verify the voltage gain of a common emitter amplifier using BJT.

THEORETICAL BACKGROUND

A Common Emitter (CE) amplifier is one of the most widely used transistor amplifier configurations because of its high voltage gain and moderate input and output impedance.

In a CE amplifier:

1. Input signal is applied between base and emitter.
2. Output signal is taken from collector and emitter.
3. The output is 180° out of phase with the input signal.
4. Voltage gain is relatively high.

At mid frequencies, coupling and bypass capacitors act as short circuits and transistor junction capacitances have negligible effect. Hence maximum gain occurs in the midband region.

DESIGN FORMULAE

Voltage Gain

$$A_v = \frac{V_o}{V_i}$$

Gain in Decibels

$$Gain(dB) = 20\log_{10}\left(\frac{V_o}{V_i}\right)$$

For CE amplifier,

$$A_v = \frac{R_C}{r_e}$$

or

$$A_v = \frac{R_C \parallel R_L}{r_e}$$

where

$$R_C \parallel R_L = \frac{R_C R_L}{R_C + R_L}$$

2. AC Emitter Resistance

$$r_e = \frac{26mV}{I_E}$$

or

$$r_e = \frac{0.026}{I_E}$$

3. Collector Current

$$I_C = \beta I_B$$

4. Base Current

$$I_B = \frac{I_C}{\beta}$$

5. Emitter Current

$$I_E = I_C + I_B$$

Approximation:

$$I_E \approx I_C$$

6. Collector Resistor

$$R_C = \frac{V_{CC} - V_{CE}}{I_C}$$

For maximum symmetrical swing:

$$V_{CEQ} = \frac{V_{CC}}{2}$$

Thus,

$$R_C = \frac{V_{CC} - V_{CEQ}}{I_C}$$

7. Emitter Resistor

$$R_E = \frac{V_E}{I_E}$$

8. Base Voltage

$$V_B = V_E + V_{BE}$$

9. Voltage Divider Bias

Upper Resistor

$$R_1 = \frac{V_{CC} - V_B}{I_{R1}}$$

Lower Resistor

$$R_2 = \frac{V_B}{I_{R2}}$$

10. Input Resistance

$$R_{in} = R_B \parallel (\beta r_e)$$

where

$$R_B = R_1 \parallel R_2$$

and

$$R_1 \parallel R_2 = \frac{R_1 R_2}{R_1 + R_2}$$

11. Output Resistance

$$R_{out} \approx R_C$$

or

$$R_{out} = R_C \parallel R_L$$

12. Load Resistance Combination

$$R_{AC} = R_C \parallel R_L$$

$$R_{AC} = \frac{R_C R_L}{R_C + R_L}$$

13. Power Dissipation

$$P_C = V_{CE} I_C$$

14. Q-Point

$$Q(V_{CEQ}, I_{CQ})$$

For Class-A operation:

$$V_{CEQ} = \frac{V_{CC}}{2}$$

$$I_{CQ} = I_C$$

Procedure for Common Emitter Amplifier Circuit using BJT:

1. Connect the circuit as per the circuit diagram which is shown above.
2. Initially connected the probe across the function generator as per shown in the circuit diagram to set the input signal.
3. Switched ON the CRO and function generator.
4. Applied the input signal as sine wave form having the valuesof5mVp-p,1KHz. from the function generator by observing in the CRO.
5. Removed the probe from that place and connected it across the C2 to observe the output of CE amplifier.

6. Switched ON the RPS and kept the +5V as VCC.

7. Kept the amplitude of the input signal as constant as 5 mV p-p for all frequency steps.

8. Noted down the values of output voltage in terms of peak to peak voltages by varying the different frequency steps in the function generator which are given below,

20 Hz, 100 Hz, 200 Hz, 500 Hz, 1 KHz, 100 KHz, 200 KHz, 400 KHz, 940 KHz, 940 KHz, 1 MHz.

9. The above readings noted in the tabular form of CE amplifier.

10. Disconnected the probe from C2 and reconnected it across C4 to observe the output of second stage.

11. Repeated the same procedure from the step 6 to 8 for tabular form of CE-CC multistage amplifier.

12. Now calculated and noted down the values in the tabular form of CE amplifier as per given below,

a). Voltage gain (Av) = Vo / Vi and Gain in dB = 20 log10 (Av).

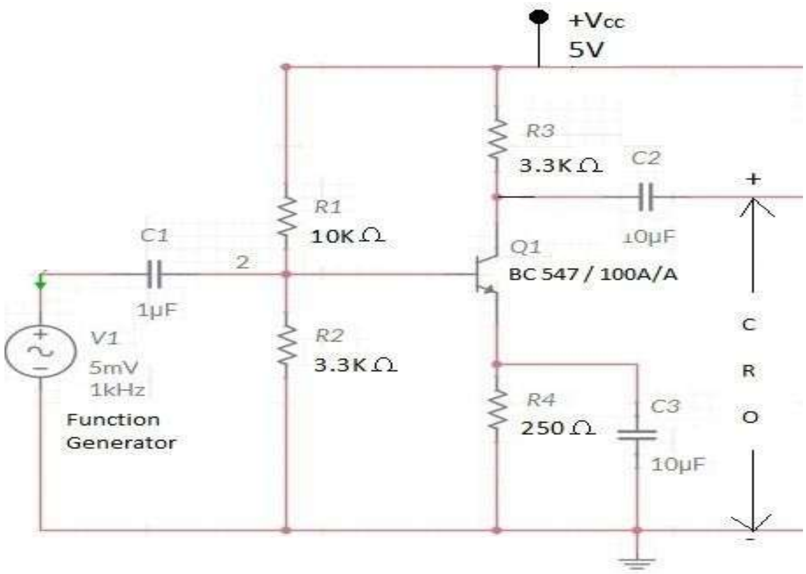
b). Plotted the graph between frequency on X-axis and gain in dB on Y-axis.

c). Band width from the graph by using the formula - Band width = f2 – f1

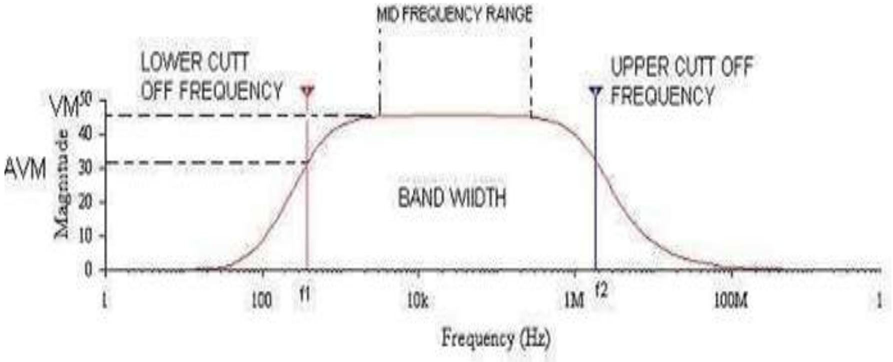
13. Now calculated and noted down the values in the tabular form of CE-CC multistage amplifier as per given below,

a). Voltage gain (Av) = Vo / Vi and Gain in dB = 20 log10 (Av).

Circuit diagram:



Model Graph:



Tabulation:

Input Voltage: $\mathbf{V_{in}}$**= 20 mV**

S.no	Frequency(Hz)	Output Voltage(V)	Gain G=V_o/Vin	Gain in dB=20 log Vo/Vin

RESULT

The midband voltage gain of the common emitter amplifier was verified successfully.

EXPERIMENT – 4.2

DETERMINATION OF LOWER AND UPPER CUTOFF FREQUENCIES FOR COMMON EMITTER AMPLIFIER

Expected Learning Outcomes

After completion of this experiment, students will be able to:

1. Identify cutoff frequencies.
2. Calculate bandwidth.
3. Understand frequency limitations of amplifiers.
4. Interpret gain-frequency characteristics.

Aim

To determine the lower and upper cutoff frequencies of a CE amplifier.

THEORETICAL BACKGROUND

At low frequencies:

Coupling capacitor reactance increases.

Emitter bypass capacitor becomes ineffective.

Therefore gain decreases.

At high frequencies:

Internal transistor capacitances dominate.

Gain decreases due to capacitive effects.

Cutoff frequencies are defined at the points where gain becomes:

$$0.707A_{v(max)}$$

or

$$-3dB$$

from the maximum gain.

DESIGN FORMULAE

Lower Cutoff Frequency

$$f_L$$

Upper Cutoff Frequency

$$f_H$$

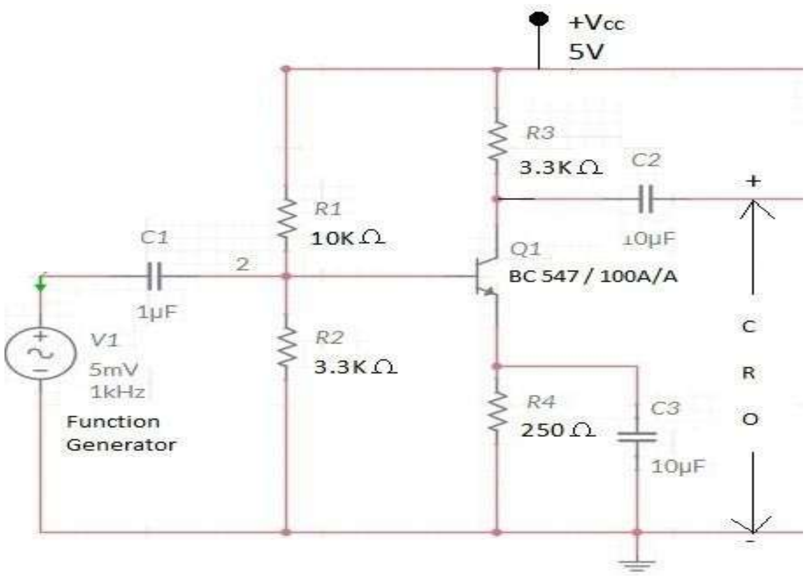
Bandwidth

$$BW = f_H - f_L$$

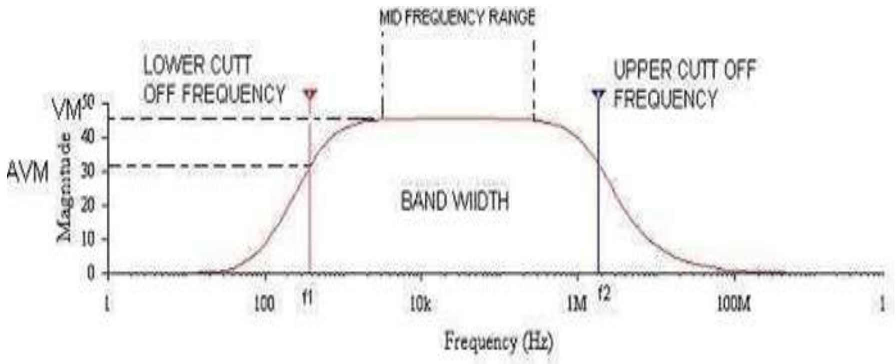
EXPERIMENTAL PROCEDURE

1. Set input voltage constant.
2. Start at 20 Hz.
3. Measure output voltage.
4. Increase frequency gradually.
5. Determine gain at each frequency.
6. Find frequencies corresponding to -3 dB points.
7. Calculate bandwidth.

Circuit diagram:



Model Graph:



Tabulation:

Input Voltage: **V_{in}= mV**

S.no	**Frequency(Hz)**	**Output Voltage(V)**	**Gain $G=V_o/Vin$**	**Gain in dB=20 log Vo/Vin**

DESIGN VERIFICATION TABLE

Parameter	**Experimental Value**
LOWER CUTOFF FREQUENCY	
UPPER CUTOFF FREQUENCY	

RESULT

The lower and upper cutoff frequencies of the CE amplifier were determined successfully.

EXPERIMENT – 4.3

BANDWIDTH ANALYSIS OF COMMON EMITTER AMPLIFIER

Expected Learning Outcomes

After completion of this experiment, students will be able to:

1. Plot frequency response characteristics.
2. Calculate bandwidth.
3. Analyze amplifier performance.
4. Construct Bode plots.

Aim

To obtain the frequency response characteristics of a CE amplifier and determine its bandwidth.

THEORY

The frequency response of an amplifier describes variation of gain with frequency.

Three frequency regions exist:

Low Frequency Region

Gain decreases due to coupling capacitors.

Mid Frequency Region

Gain remains approximately constant.

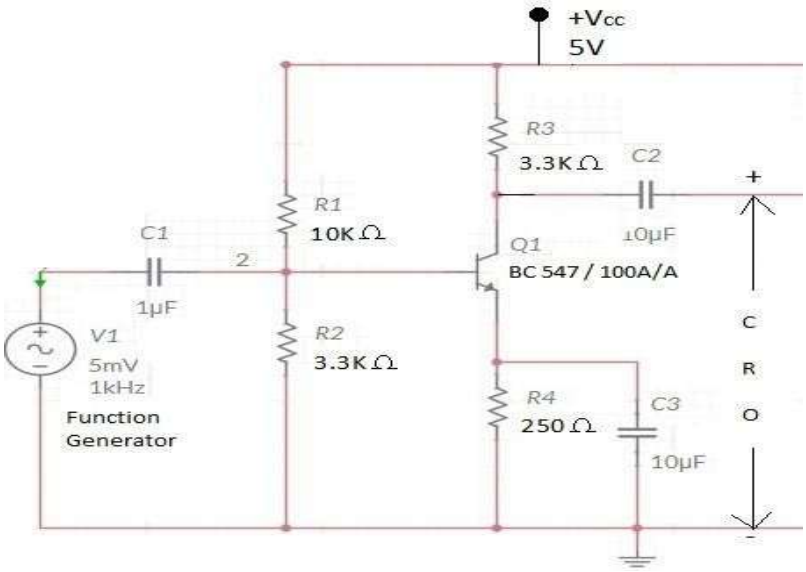
High Frequency Region

Gain decreases because of transistor junction capacitances.

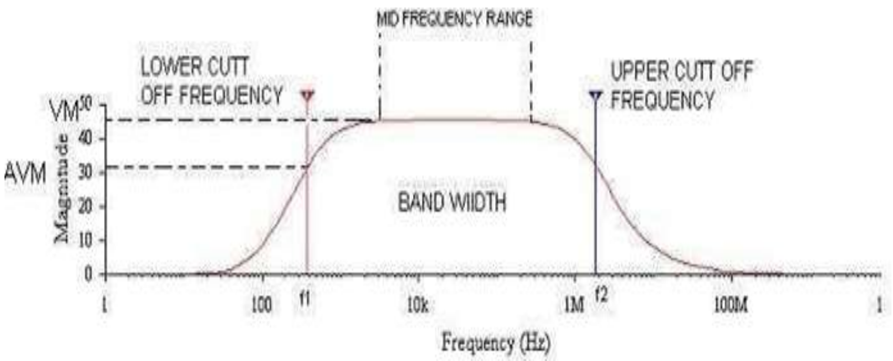
EXPERIMENTAL PROCEDURE

1. Apply constant input voltage.
2. Start at 20 Hz.
3. Record output voltage.
4. Increase frequency gradually.
5. Continue measurements up to 1 MHz.
6. Calculate gain.
7. Convert gain into dB.
8. Plot Bode magnitude response.
9. Determine bandwidth.

Circuit diagram:



Model Graph:



Tabulation:

Input Voltage: $\mathbf{V_{in}}$**= 20 mV**

S.no	Frequency(Hz)	Output Voltage(V)	Gain $G=V_o/V_{in}$	Gain in dB=20 log Vo/Vin

DESIGN VERIFICATION TABLE

Parameter	Experimental Value
LOWER CUTOFF FREQUENCY	
UPPER CUTOFF FREQUENCY	
BANDWIDTH	

RESULT

The frequency response characteristics of the common emitter amplifier were obtained successfully and the bandwidth was determined.

EXPERIMENT – 5

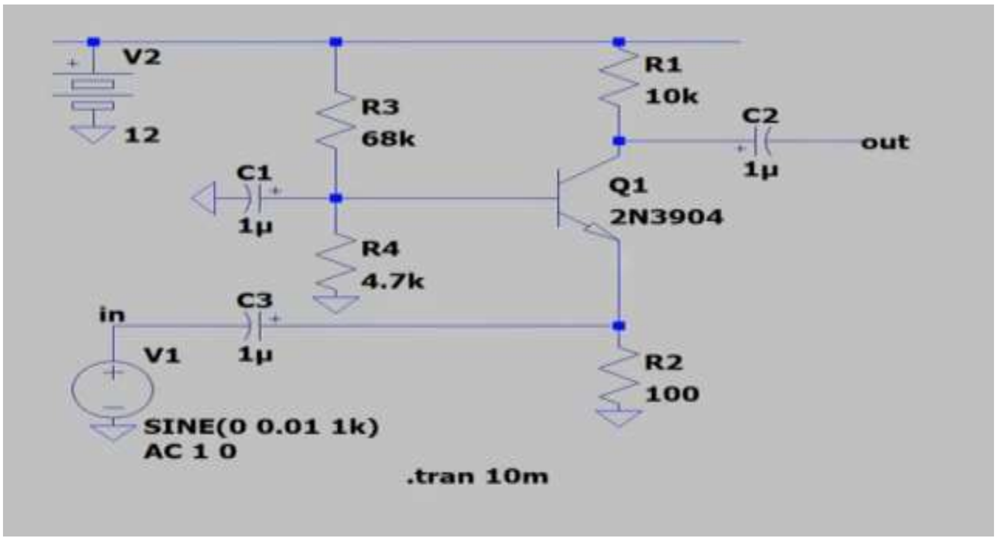
DESIGN AND SIMULATION OF COMMON BASE AMPLIFIER USING LTSPICE SIMULATOR

OBJECTIVES

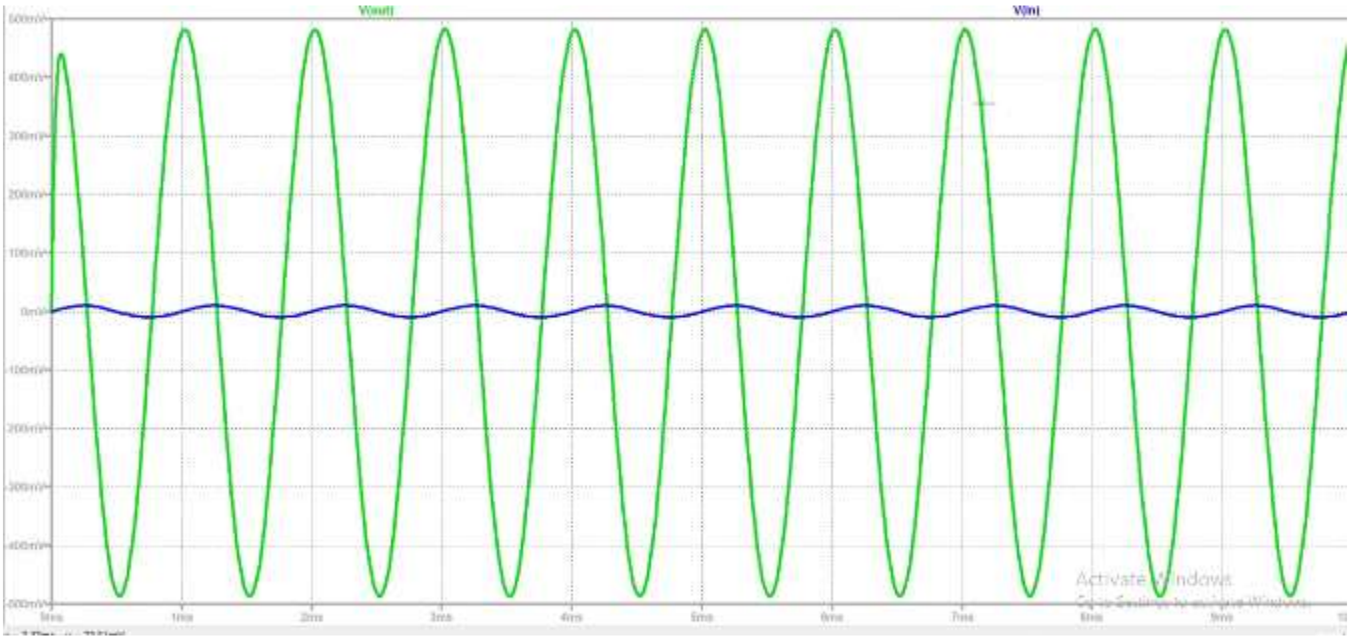
The experiment is divided into three engineering laboratory model tests:

1. Analysis of Voltage Gain
2. Input and Output Characteristic Analysis
3. Frequency Response Analysis

Circuit Diagram:



Model Graph:



EXPERIMENT – 5.1

VOLTAGE GAIN ANALYSIS OF COMMON BASE AMPLIFIER USING LTSPICE

Expected Learning Outcomes

After completion of this experiment, students will be able to:

1. Design a Common Base amplifier using LTSpice.
2. Simulate amplifier operation.
3. Measure voltage gain.
4. Compare theoretical and simulated results.
5. Analyze CB amplifier characteristics.

Aim

To design and simulate a Common Base amplifier using LTSpice and verify its voltage gain.

THEORETICAL BACKGROUND

The **Common Base (CB) amplifier** is one of the three basic transistor amplifier configurations in which the **base terminal is common** to both the input and output circuits. In this configuration, the input signal is applied between the emitter and base terminals, while the output is taken between the collector and base terminals.

The Common Base amplifier is characterized by:

- Low input resistance
- High output resistance
- High voltage gain
- Current gain less than unity
- Excellent high-frequency performance

The CB amplifier is widely used in:

- RF amplifiers
- Impedance matching circuits
- High-frequency applications

DESIGN FORMULA:

Voltage Gain

$$A_v = \frac{V_o}{V_i}$$

Current Gain

$$A_i = \alpha$$

where

$$\alpha = \frac{I_C}{I_E}$$

EXPERIMENTAL PROCEDURE

1. Open LTSpice.
2. Create a new schematic.
3. Place NPN transistor.
4. Add biasing resistors.
5. Configure Common Base circuit.
6. Apply AC signal source.
7. Set simulation parameters.
8. Run transient analysis.
9. Measure input and output voltages.
10. Calculate gain.
11. Record observations.

TABULATION:

s.no	INPUT VOLTAGE Vi (mV)	OUTPUT VOLTAGE V0 (V)	GAIN Av

RESULT:

The voltage gain of the Common Base amplifier was verified successfully using LTSpice

EXPERIMENT – 5.2

MEASUREMENT OF INPUT AND OUTPUT CHARACTERISTICS OF COMMON BASE AMPLIFIER USING LTSPICE

Expected Learning Outcomes

After completion of this experiment, students will be able to:

1. Plot input characteristics.
2. Plot output characteristics.
3. Analyze transistor operation.

Aim

To obtain the input and output characteristics of a Common Base amplifier using LTSpice.

EXPERIMENTAL PROCEDURE

1. Configure DC sweep analysis.
2. Vary emitter voltage.
3. Measure emitter current.
4. Plot input characteristics.
5. Vary collector voltage.
6. Measure collector current.
7. Plot output characteristics.
8. Determine resistances.

EXPERIMENTAL OBSERVATIONS

INPUT CHARECTERISTICS

Parameter	PRACTICAL VALUE
AMPLITURE	
TIME PERIOD	

OUTPUT CHARECTERISTICS

Parameter	PRACTICAL VALUE
AMPLITURE	
TIME PERIOD	

RESULT

The input and output characteristics of the Common Base amplifier were obtained successfully.

EXPERIMENT – 5.3

FREQUENCY RESPONSE ANALYSIS COMMON BASE AMPLIFIER USING LTSPICE

The experiment is divided into three engineering laboratory model tests:

Expected Learning Outcomes

After completion of this experiment, students will be able to:

1. Perform AC analysis.
2. Plot frequency response.
3. Determine bandwidth.

Aim

To obtain the frequency response characteristics of a Common Base amplifier using LTSpice.

THEORY

The Common Base amplifier exhibits excellent high-frequency response due to:

- Absence of Miller effect.
- Low input capacitance.
- Reduced feedback capacitance.

Therefore, CB amplifiers are preferred for RF applications.

DESIGN FORMULA:

Voltage Gain

$$A_v = \frac{V_o}{V_i}$$

Gain in dB

$$Gain(dB) = 20\log_{10}(A_v)$$

Bandwidth

$$BW = f_H - f_L$$

EXPERIMENTAL PROCEDURE

1. Configure AC analysis.
2. Sweep frequency from 10 Hz to 100 MHz.
3. Observe gain.
4. Plot Bode response.
5. Determine cutoff frequencies.
6. Calculate bandwidth.
7. Record observations.

BANDWIDTH

PARAMETER	VOUT
AMPLITUDE	
TIME PERIOD	
BANDWIDTH	

RESULT

The frequency response characteristics of the Common Base amplifier were analyzed successfully using LTSpice.

EXPERIMENT – 6

DESIGN AND ANALYSIS OF ASTABLE MULTIVIBRATOR USING BJT

OBJECTIVES

The experiment is divided into three engineering laboratory model tests:

1. Switching Analysis Astable Multivibrator
2. Measurement of Time Period and Frequency
3. Duty Cycle Analysis

EXPERIMENT – 6.1

SWITCHING ANALYSIS OF ASTABLE MULTIVIBRATOR

Expected Learning Outcomes

After completion of this experiment, students will be able to:

1. Design an astable multivibrator using BJT.
2. Observe continuous switching operation.
3. Analyze transistor switching behavior.
4. Generate square wave signals.
5. Compare theoretical and practical waveforms.

Aim

To design and verify the operation of an astable multivibrator using BJT.

THEORETICAL BACKGROUND

An astable multivibrator is a free-running oscillator having no stable state. The circuit continuously switches between two quasi-stable states due to regenerative feedback.

Characteristics:

- No external triggering required.
- Generates square wave output.
- Used as clock pulse generator.
- Produces continuous oscillations.

During operation:

- One transistor conducts while the other remains OFF.
- Capacitors charge and discharge alternately.
- Switching occurs automatically.

DESIGN FORMULA:

Time Period

$$T = 1.38RC$$

Frequency

$$f = \frac{1}{T}$$

ON Time

$$T_1 = 0.69R_1C_1$$

OFF Time

$$T_2 = 0.69R_2C_2$$

2. Frequency of Oscillation

$$f = \frac{1}{T}$$

Substituting T:

$$f = \frac{1}{1.386RC}$$

Approximation:

$$f \approx \frac{0.72}{RC}$$

3. Capacitor Calculation

$$C = \frac{1}{1.386fR}$$

4. Resistor Calculation

$$R = \frac{1}{1.386fC}$$

5. ON Time of Transistor Q1

$$t_{ON1} = 0.693R_2C_1$$

6. ON Time of Transistor Q2

$$t_{ON2} = 0.693R_3C_2$$

7. Total Time Period

$$T = t_{ON1} + t_{ON2}$$

$$T = 0.693(R_2C_1 + R_3C_2)$$

8. Frequency (General Case)

$$f = \frac{1}{0.693(R_2C_1 + R_3C_2)}$$

9. Duty Cycle

$$D = \frac{t_{ON}}{T} \times 100$$

For symmetrical operation:

$$D = 50\%$$

when

$$R_2 = R_3$$

and

$$C_1 = C_2$$

10. Collector Current

$$I_C = \beta I_B$$

11. Base Current

$$I_B = \frac{I_C}{\beta}$$

12. Base Resistor

$$R_B = \frac{V_{CC} - V_{BE}}{I_B}$$

13. Collector Resistor

$$R_C = \frac{V_{CC} - V_{CE(sat)}}{I_C}$$

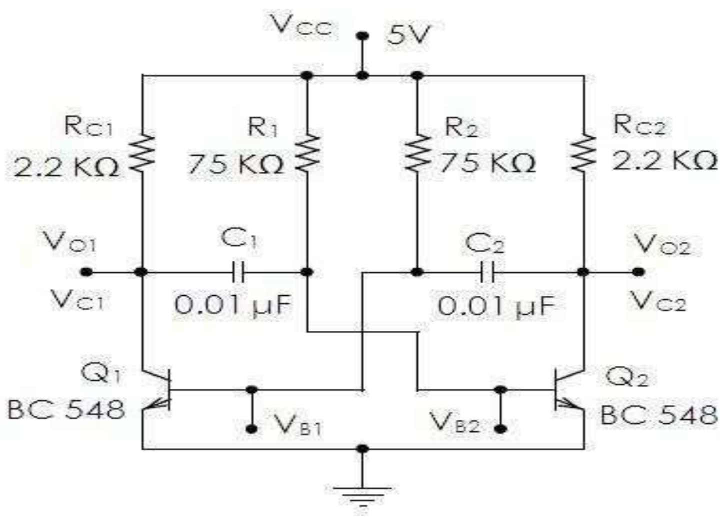
14. Output Peak-to-Peak Voltage

$$V_{pp} \approx V_{CC}$$

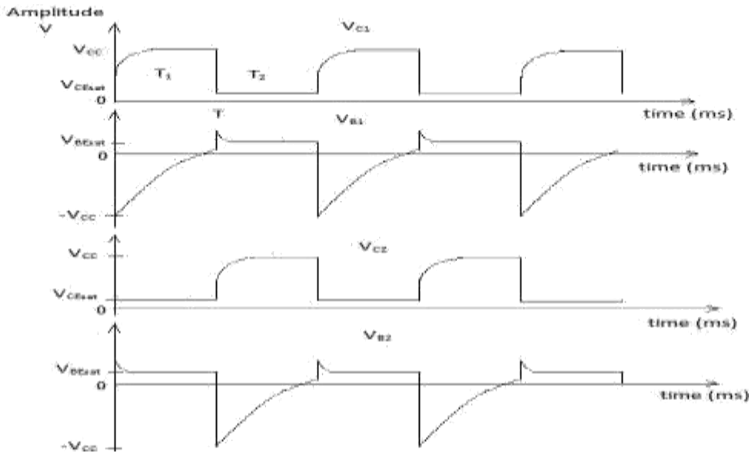
EXPERIMENTAL PROCEDURE

1. Verify all components.
2. Assemble the astable multivibrator circuit.
3. Apply DC supply voltage.
4. Observe collector and base outputs of both transistors.
5. Verify alternate switching.
6. Observe square wave output.
7. Measure time period.
8. Record waveform characteristics.

CIRCUIT DIAGRAM:



MODEL GRAPH



Tabulation:

S.no	Amplitude (Volts)	TON (ms)	TOFF (ms)
Theoretical value			
Practical Value			

Result:

Astable Multivibrator was designed and frequency of oscillation was calculated.

Frequency of oscillation f= ________(Hz)

EXPERIMENT – 6.2

MEASUREMENT OF TIME PERIOD AND FREQUENCY OF ASTABLE MULTIVIBRATOR

Expected Learning Outcomes

After completion of this experiment, students will be able to:

1. Measure time period experimentally.
2. Determine oscillation frequency.
3. Verify theoretical calculations.
4. Analyze capacitor charging and discharging effects.

Aim

To measure the time period and frequency of an astable multivibrator.

THEORETICAL BACKGROUND

The oscillation frequency depends on resistor and capacitor values. Capacitors alternately charge and discharge through base resistors causing periodic switching.

For symmetrical operation:

$$T = 1.38RC$$

Frequency:

$$f = \frac{1}{T}$$

Increasing R or C increases time period and decreases frequency.

Time Period $T=T_1+T_2$ where T_1= ON time T_2= OFF time

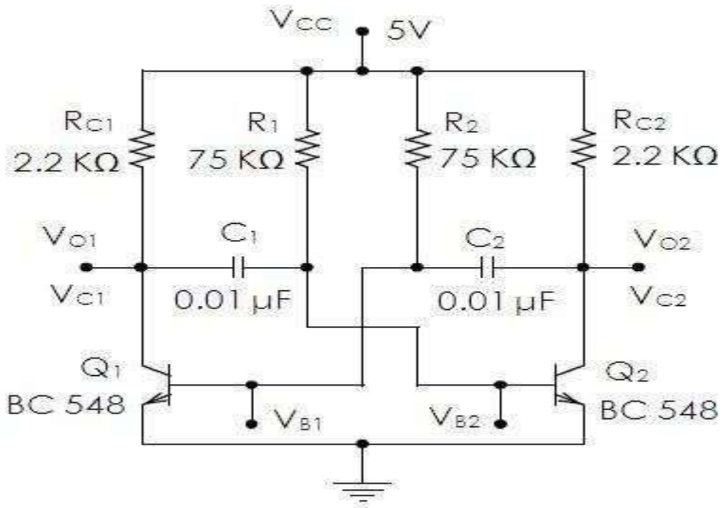
Frequency f=1/ T in Hz.

EXPERIMENTAL PROCEDURE

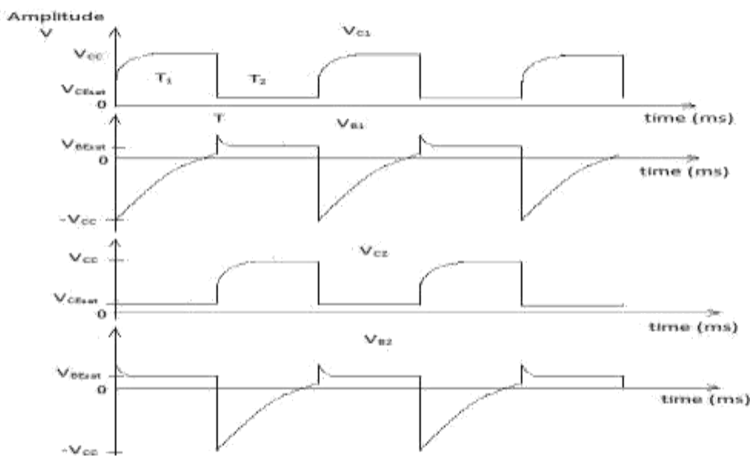
1. Assemble the astable multivibrator.
2. Measure waveform using CRO.
3. Determine one complete cycle.
4. Measure time period.
5. Calculate frequency.

6. Compare theoretical and practical values.
7. Record observations.

Circuit diagram:



Model Graph:



Procedure:

1. Test the components.
2. Assemble the circuit on a breadboard.
3. Switch ON the power supply.
4. Connect the outputs of the circuit to an oscilloscope.
5. Observe the **collector** and **base** waveforms of the two transistors.
6. Measure the **frequency** and **amplitude** of the outputs.
7. Plot the waveforms.

Tabulation:

S.no	Amplitude (Volts)	T_{ON} (ms)	T_{OFF} (ms)	Frequency (Hz)
Theoretical value				
Practical Value				

RESULT

The time period and frequency of the astable multivibrator were measured successfully.

EXPERIMENT – 6.3

DUTY CYCLE ANALYSIS OF ASTABLE MULTIVIBRATOR

Expected Learning Outcomes

After completion of this experiment, students will be able to:

1. Measure ON time and OFF time.
2. Calculate duty cycle.
3. Analyze waveform symmetry.
4. Evaluate pulse generation characteristics.

Aim

To determine the duty cycle of the astable multivibrator.

THEORY

Duty cycle is defined as the ratio of ON time to total time period.

Duty Cycle

$$D = \frac{T_{ON}}{T} \times 100$$

where

$$T = T_{ON} + T_{OFF}$$

For symmetrical astable multivibrators:

$$D \approx 50\%$$

DESIGN FORMULA:

ON Time

$$T_{ON} = 0.69RC$$

OFF Time

$$T_{OFF} = 0.69RC$$

Duty Cycle

$$D = \frac{T_{ON}}{T_{ON} + T_{OFF}} \times 100$$

EXPERIMENTAL PROCEDURE

1. Observe output waveform on CRO.
2. Measure ON duration.
3. Measure OFF duration.
4. Calculate total period.
5. Determine duty cycle.
6. Compare with theoretical value.
7. Record observations.

Duty Cycle Measurement

Test Case	TON (ms)	TOFF (ms)	Time Period T (ms)	Duty Cycle (%)

RESULT

The duty cycle of the astable multivibrator was determined successfully and verified with theoretical calculations.

EXPERIMENT – 7

DESIGN AND ANALYSIS OF SCHMITT TRIGGER USING LTSPICE

OBJECTIVES

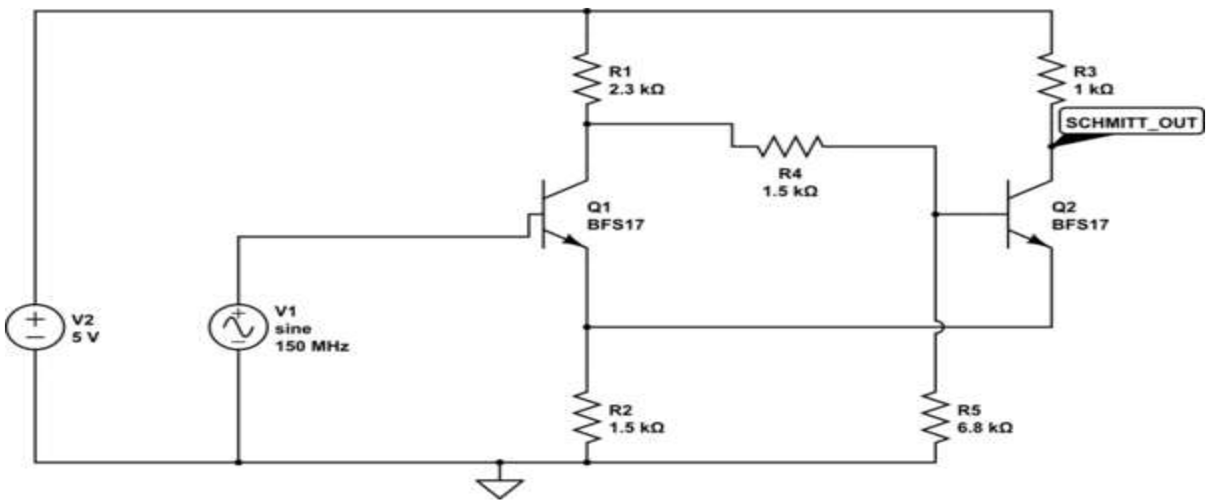
The experiment is divided into three engineering laboratory model tests:

1.Analyse function of Schmitt trigger and determine bandwidth

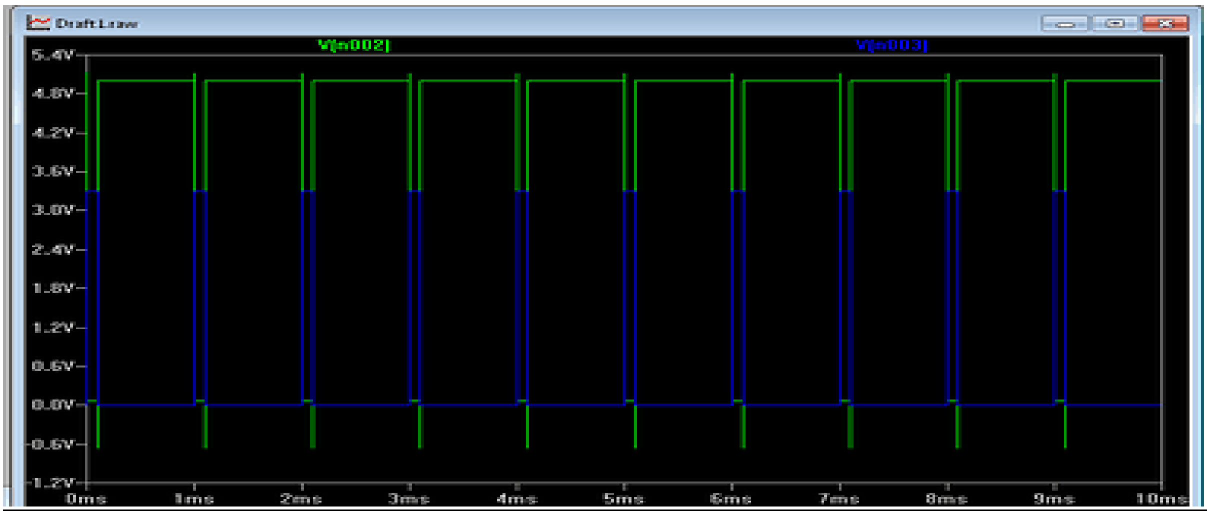
2.Determination Of UTP and LTP Frequency Response Analysis

3.Analysis of Hysteresis Characteristics

CIRCUIT DIAGRAM:



<u>Model Graph:</u>



EXPERIMENT – 7.1

BANDWIDTH ANALYSIS OF SCHMITT TRIGGER USING LTSPICE

Expected Learning Outcomes

After completion of this experiment, students will be able to:

- Understand regenerative feedback.
- Analyze bistable switching action.
- Measure trigger levels.
- Determine bandwidth

Aim

To simulate and verify the operation of a Schmitt Trigger and determine bandwidth using LTSpice.

THEORY

A Schmitt Trigger is a regenerative comparator circuit exhibiting hysteresis.

It has:

- Two stable states.
- Two switching thresholds.
- Excellent noise immunity.

Procedure:

1.Connections are made as shown in the circuit diagram.

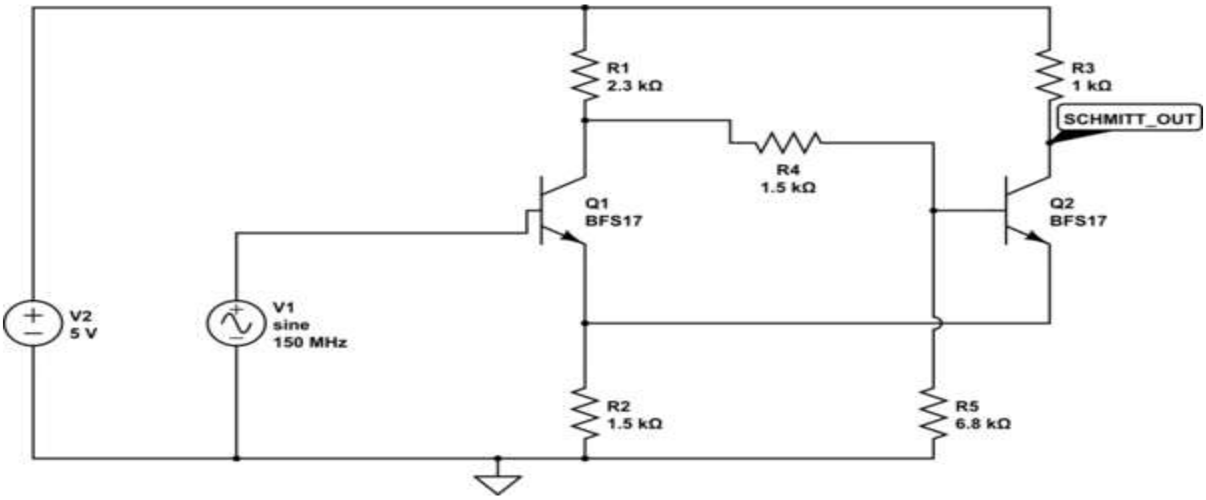
2. The DC power supply is switched ON.

3. The output waveform is displayed on the CRO.

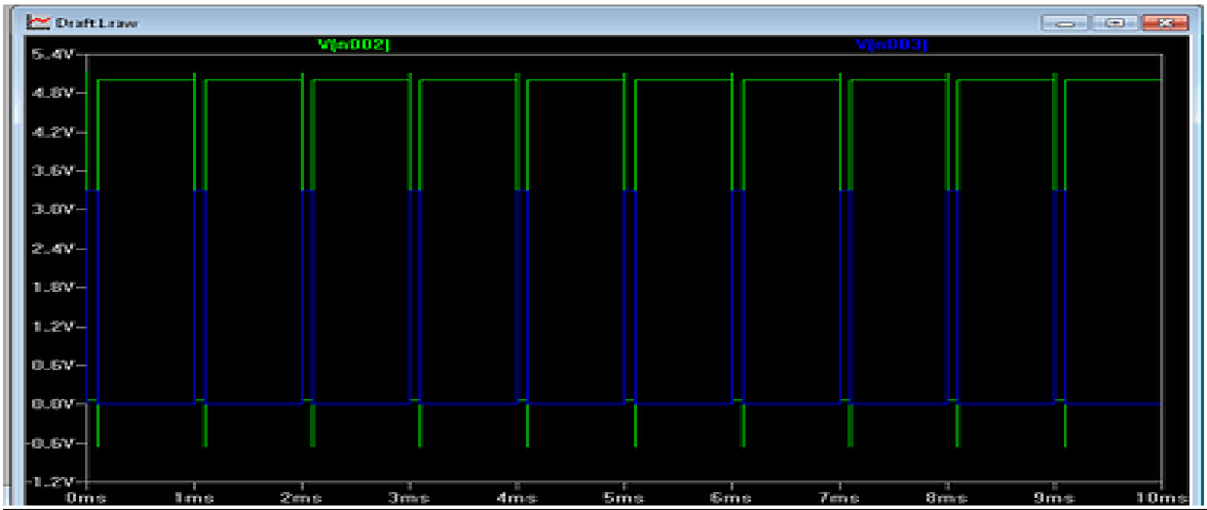
4. The peak-to-peak Amplitude and time period of the sine wave is noted.

5. The graph of output waveform is drawn.

CIRCUIT DIAGRAM:



Model Graph:



BANDWIDTH

PARAMETER	VOUT
AMPLITUDE	
TIME PERIOD	
BANDWIDTH	

Result:

Schmitt Trigger using LTSpice was completed and output was obtained.

EXPERIMENT – 7.2

DETERMINATION OF UTP AND LTP FOR SCHMITT TRIGGER USING LTSPICE

After completion of this experiment, students will be able to:

1. Understand the operating principle of a Schmitt Trigger circuit.
2. Explain the concept of hysteresis and its significance in switching circuits.
3. Determine the **Upper Trigger Point (UTP)** and **Lower Trigger Point (LTP)** experimentally.
4. Calculate the hysteresis width of the Schmitt Trigger circuit.
5. Analyze the effect of positive feedback on circuit operation.
6. Apply Schmitt Trigger circuits in waveform shaping and signal conditioning applications.

Aim

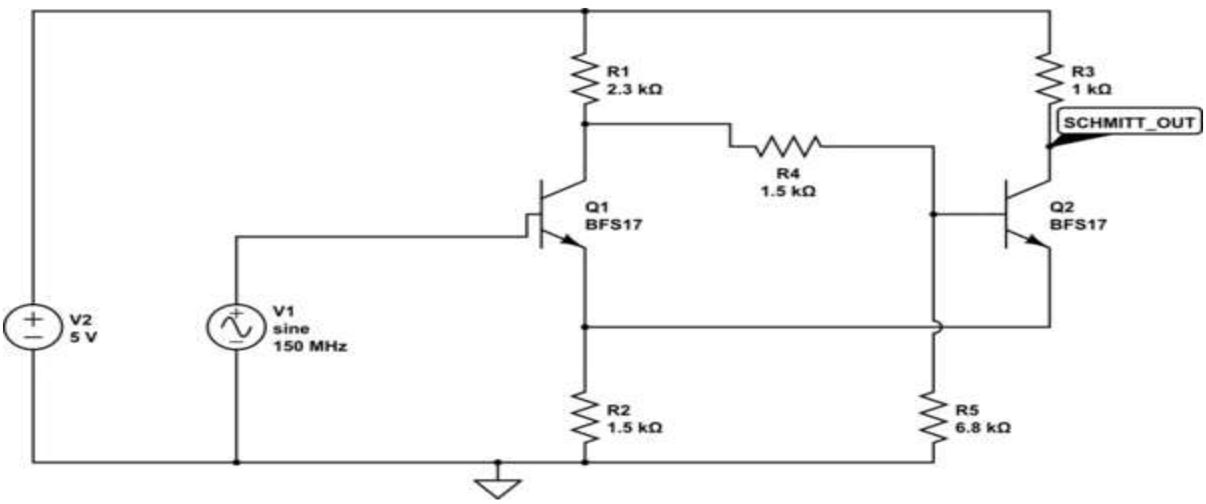
To determine Upper Trigger Point and Lower Trigger Point for Schmit trigger circuit

DESIGN FORMULA:

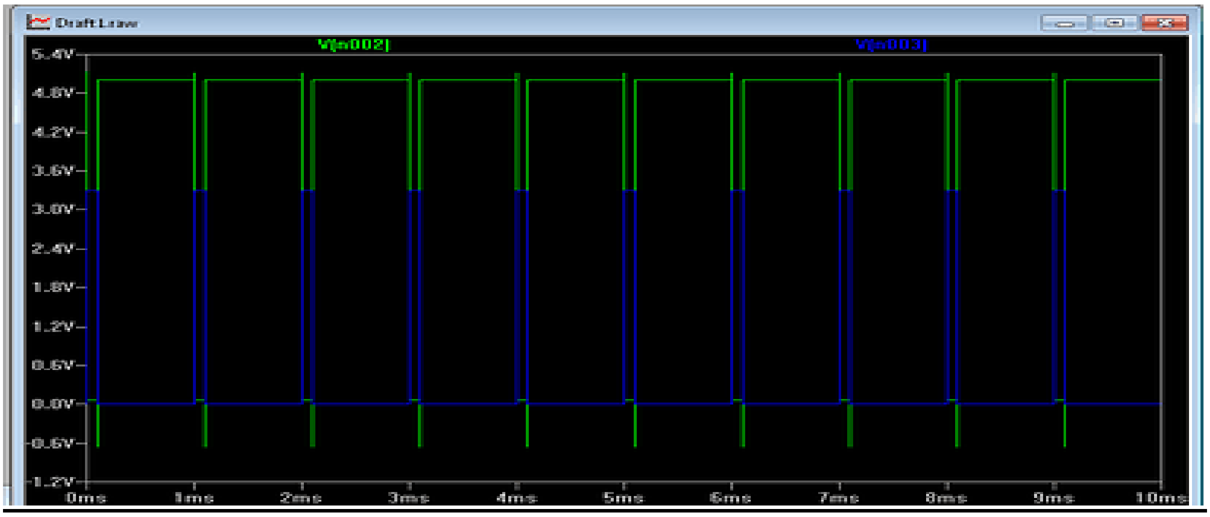
Hysteresis Width

$$V_H = UTP - LTP$$

CIRCUIT DIAGRAM:



Model Graph:



TABULATION

PARAMETER	VOUT
UTP	
LTP	
HYSTERESIS WIDTH	

RESULT

The trigger levels were measured successfully.

EXPERIMENT – 7.3

ANALYSIS OF HYSTERESIS CHARACTERISTICS FOR SCHMITT TRIGGER USING LTSPICE

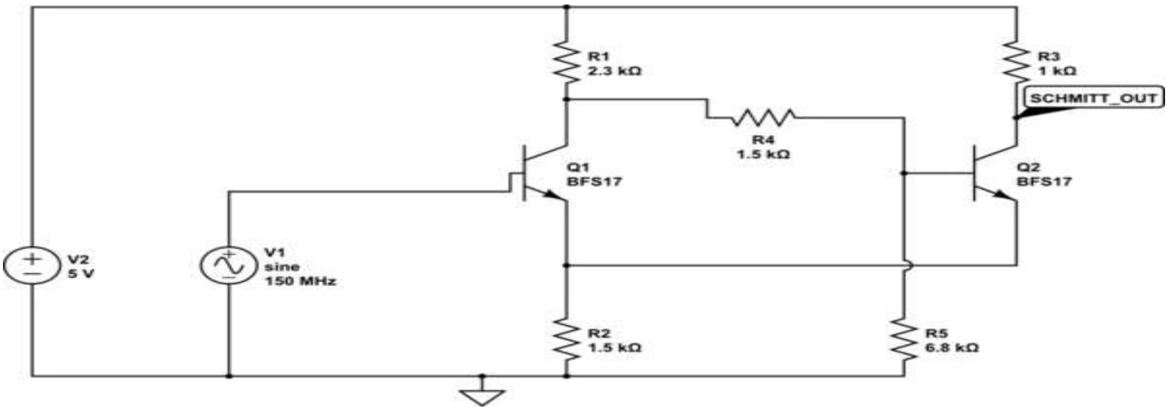
After completion of this experiment, students will be able to:

1. Understand the operating principle of a Schmitt Trigger circuit.
2. Explain the concept of hysteresis and its significance in switching circuits.
3. Determine the **Upper Trigger Point (UTP)** and **Lower Trigger Point (LTP)** experimentally.
4. Calculate the hysteresis width of the Schmitt Trigger circuit.
5. Analyze the effect of positive feedback on circuit operation.
6. Observe and interpret the switching characteristics of the Schmitt Trigger.
7. Plot and analyze the hysteresis transfer characteristic curve.
8. Evaluate the noise immunity provided by hysteresis in comparator circuits.
9. Compare the operation of a Schmitt Trigger with that of a conventional comparator.
10. Apply Schmitt Trigger circuits in waveform shaping and signal conditioning applications.

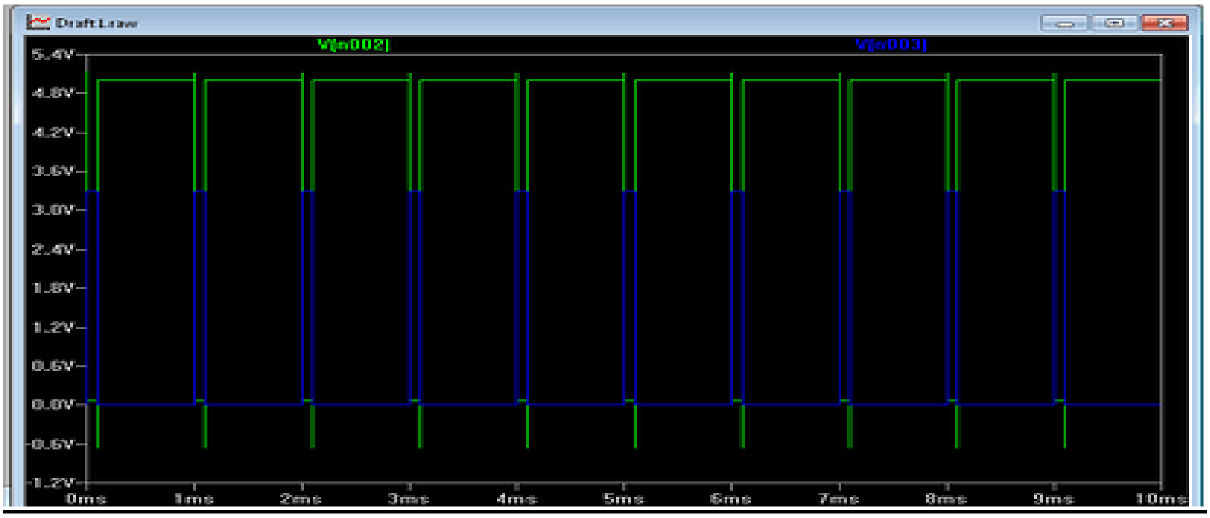
Aim

To plot hysteresis characteristics of a Schmitt Trigger.

CIRCUIT DIAGRAM:



Model Graph:



TABULATION

PARAMETER	VOUT
VIN	
VOUT	

RESULT:

Thus the plot hysteresis characteristics of a Schmitt Trigger was done.

EXPERIMENT – 8

DESIGN AND SIMULATION OF UJT RELAXATION OSCILLATOR USING LTSPICE

OBJECTIVES

1. Analysis of UJT Relaxation Oscillator.
2. Determine Bandwidth
3. Determine Duty Cycle

EXPERIMENT – 8.1

FREQUENCY ANALYSIS OF UJT RELAXATION OSCILLATOR USING LTSPICE

After completion of this experiment, students will be able to:

1. Understand the construction and operating principle of a Unijunction Transistor (UJT).
2. Explain the working of a UJT Relaxation Oscillator.
3. Verify the generation of oscillations using a UJT-based oscillator circuit.
4. Observe the charging and discharging characteristics of the timing capacitor.
5. Measure the oscillation frequency and compare it with the theoretical value.
6. Analyze the effect of resistance and capacitance values on the oscillation frequency.
7. Identify and interpret the sawtooth waveform and trigger pulse waveform generated by the circuit.
8. Determine the role of the intrinsic stand-off ratio (η)(\eta)(η) in oscillator operation.
9. Plot and analyze the output waveforms using a CRO/oscilloscope or simulation tool.
10. Apply UJT oscillators in timing, triggering, pulse generation, and sweep generation applications.

Aim

To verify the operation of a UJT Relaxation Oscillator.

THEORY

The UJT relaxation oscillator generates non-sinusoidal oscillations.

Operation:

- Capacitor charges exponentially.
- UJT triggers at peak voltage.
- Capacitor discharges rapidly.
- Cycle repeats continuously.

DESIGN FORMULA:

Frequency

$$f = \frac{1}{RC \ln\left(\frac{1}{1-\eta}\right)}$$

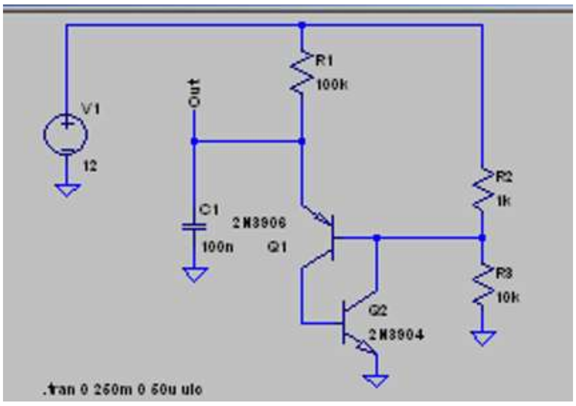
where

η = Intrinsic Stand-off Ratio

PROCEDURE

1. Draw UJT oscillator in LTSpice.
2. Configure RC network.
3. Run transient analysis.
4. Observe sawtooth waveform.
5. Measure oscillation frequency.

Circuit diagram:



Procedure:

1.Connections are made as shown in the circuit diagram.

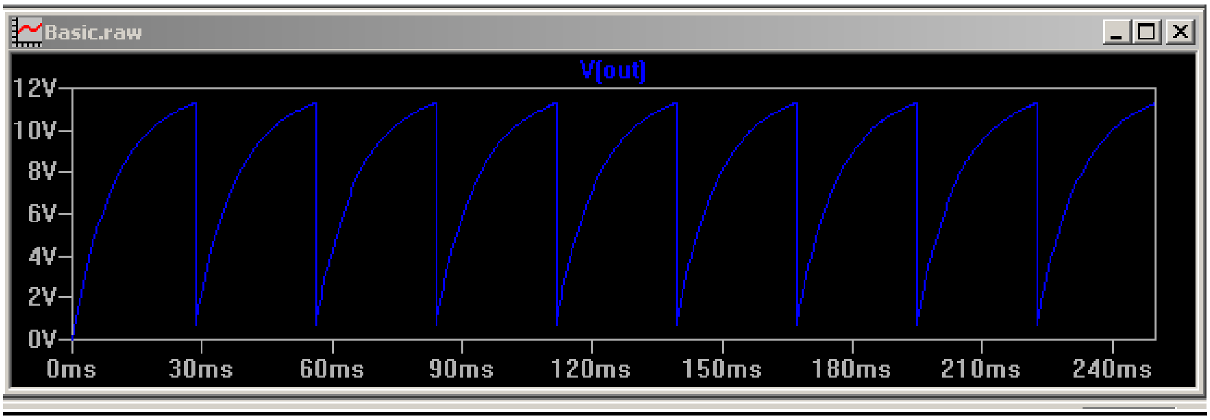
2. The DC power supply is switched ON.

3. The output waveform is displayed on the CRO.

4. The peak-to-peak Amplitude and time period of the sine wave is noted.

5. The graph of output waveform is drawn.

Model Graph:



OBSERVATION TABLE

Waveform	Observation
Amplitude	
Time period	
Frequency	

Result:

UJT Relaxation Oscillator using LTSpice was completed and output obtained.

EXPERIMENT – 8.2

BANDWIDTH ANALYSIS OF UJT RELAXATION OSCILLATOR USING LTSPICE

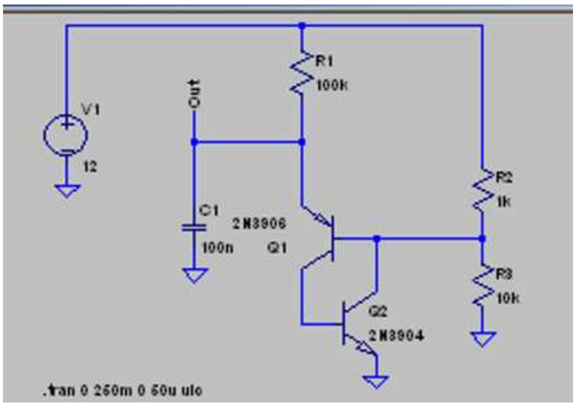
After completion of this experiment, students will be able to:

1. Understand the operating principle of a UJT Relaxation Oscillator and its frequency-determining mechanism.
2. Simulate a UJT Relaxation Oscillator using LTSpice software.
3. Analyze the charging and discharging behavior of the timing capacitor in the oscillator circuit.
4. Measure the oscillation frequency from the simulated output waveform.
5. Compare theoretical and simulated frequency values of the oscillator.
6. Investigate the effect of varying resistance (R)(R)(R) and capacitance (C)(C)(C) values on the oscillation frequency.
7. Observe and interpret the generated sawtooth and pulse waveforms using LTSpice.
8. Plot and analyze the relationship between frequency and RC time constant.
9. Apply LTSpice simulation tools for frequency analysis and performance evaluation of relaxation oscillators.

Aim

To verify frequency variation with RC values.

Circuit diagram:



Procedure:

1.Connections are made as shown in the circuit diagram.

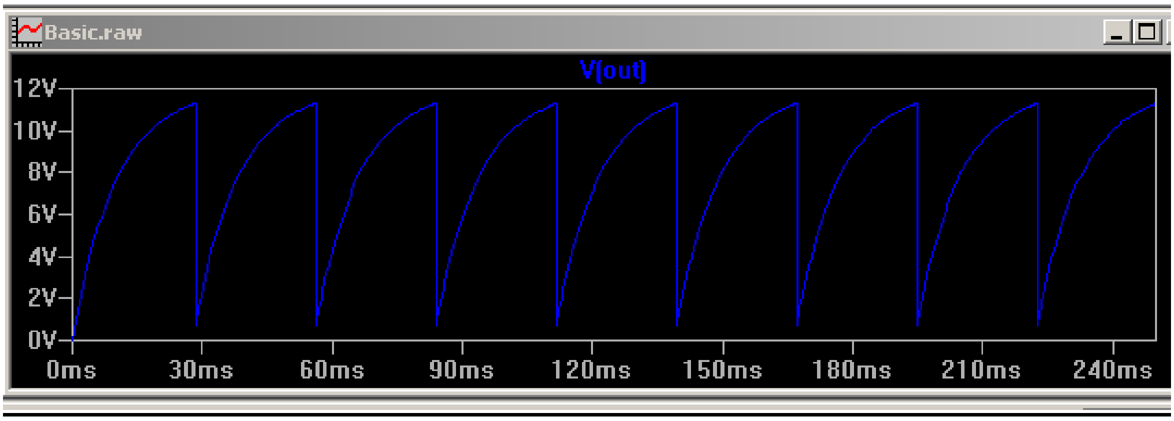
2. The DC power supply is switched ON.

3. The output waveform is displayed on the CRO.

4. The peak-to-peak Amplitude and time period of the sine wave is noted.

5. The graph of output waveform is drawn.

Model Graph:



OBSERVATION TABLE

Waveform	Observation
Amplitude	
Time period	
Frequency	
Bandwidth	

Result:

Thus the bandwidth of UJT Relaxation Oscillator using LTSpice was completed and output obtained.

EXPERIMENT – 8.3

WAVEFORM ANALYSIS OF UJT RELAXATION OSCILLATOR WITH DUTY CYCLE USING LTSPICE

After completion of this experiment, students will be able to:

1. Understand the operating principle of a UJT Relaxation Oscillator and its frequency-determining mechanism.
2. Simulate a UJT Relaxation Oscillator using LTSpice software.
3. Analyze the charging and discharging behavior of the timing capacitor in the oscillator circuit.
4. Measure the oscillation frequency from the simulated output waveform.
5. Compare theoretical and simulated frequency values of the oscillator.
6. Investigate the effect of varying resistance (R)(R)(R) and capacitance (C)(C)(C) values on the oscillation frequency.
7. Observe and interpret the generated sawtooth and pulse waveforms using LTSpice.
8. Plot and analyze the relationship between frequency and RC time constant.
9. Apply LTSpice simulation tools for frequency analysis and performance evaluation of relaxation oscillators.

EXPERIMENT – 8.3

Aim

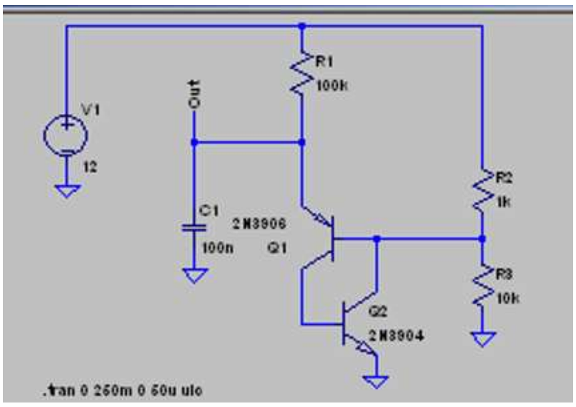
To observe generated waveforms duty cycle

Duty Cycle

- Ratio of pulse width to total period:

$$\text{Duty Cycle} = \frac{\text{Pulse Width}}{\text{Period}} \times 100\%$$

Circuit diagram:



Procedure:

1.Connections are made as shown in the circuit diagram.

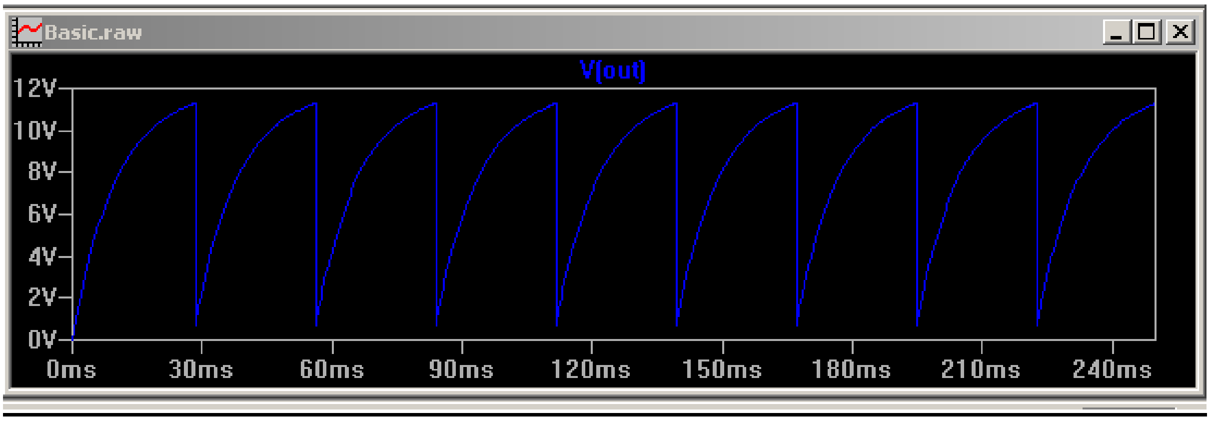
2. The DC power supply is switched ON.

3. The output waveform is displayed on the CRO.

4. The peak-to-peak Amplitude and time period of the sine wave is noted.

5. The graph of output waveform is drawn.

Model Graph:



OBSERVATION TABLE

Waveform	THEORITICAL	PRACTICAL
Pulse width		
period		
Duty Cycle		

Result:

UJT Relaxation Oscillator using LTSpice was completed and output obtained.

EXPERIMENT – 9

Design of Hartley Oscillator Using PCB Software

OBJECTIVES

1. Design and Create PCB layout for Hartley Oscillator circuit.
2. Route PCB tracks with Manufacturing board
3. Generate fabrication files.

Expected Learning Outcomes

After completion of this experiment, students will be able to:

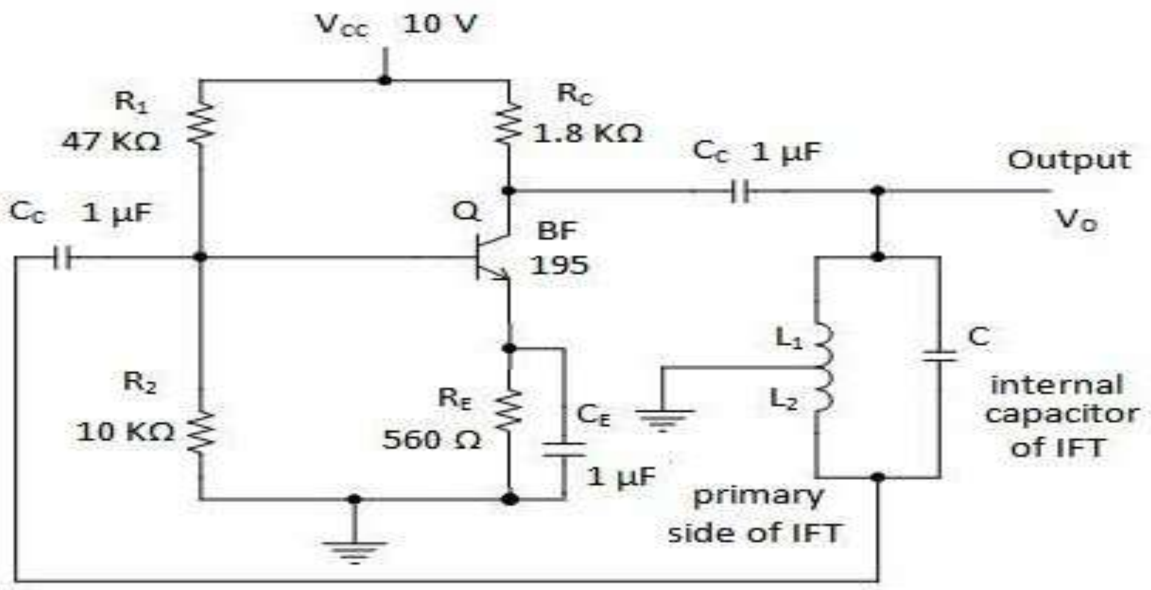
1. Understand the operating principle of a Hartley Oscillator and its applications.
2. Design the schematic of a Hartley Oscillator using EAGLE CAD software.
3. Calculate the oscillation frequency using inductance and capacitance values.
4. Create a PCB layout by placing components and routing tracks effectively.
5. Perform Design Rule Check (DRC) to verify PCB manufacturability.
6. Generate Gerber and drill files required for PCB fabrication.
7. Analyze PCB design considerations such as component placement, grounding, and signal routing.
8. Apply modern PCB design tools for analog circuit implementation.

Theory

A Hartley oscillator is an LC oscillator that uses a tapped inductor for feedback. The total inductance of the tank circuit determines the frequency of oscillation. Positive feedback is achieved through inductive coupling. The condition for sustained oscillations is given by the Barkhausen criterion.

$$f = \frac{1}{2\pi\sqrt{(L_1 + L_2)C}}$$

Circuit Diagram



EXPERIMENT 9.1

DESIGN A HARTLEY OSCILLATOR CIRCUIT USING PCB SOFTWARE AND IMPLEMENT ITS PCB LAYOUT

Aim

To design a Hartley oscillator circuit and implement its PCB layout using EAGLE software

Apparatus / Software Required

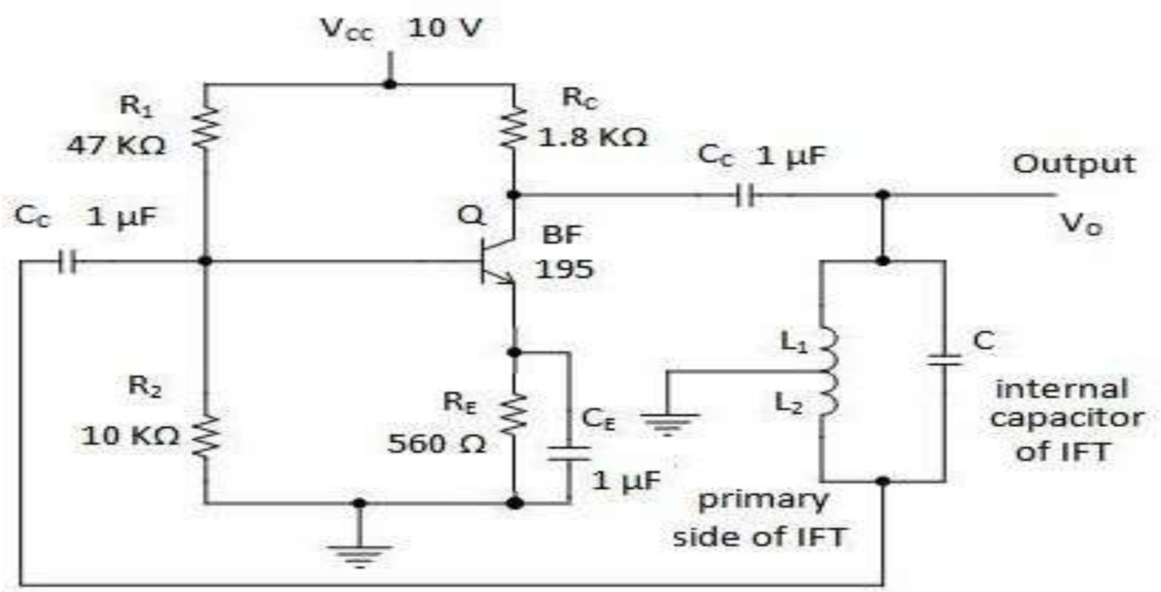
- PCB CAD Software
- PC

Theory

A Hartley oscillator is an LC oscillator that uses a tapped inductor for feedback. The total inductance of the tank circuit determines the frequency of oscillation. Positive feedback is achieved through inductive coupling. The condition for sustained oscillations is given by the Barkhausen criterion.

$$f = \frac{1}{2\pi\sqrt{(L_1 + L_2)C}}$$

Circuit diagram:

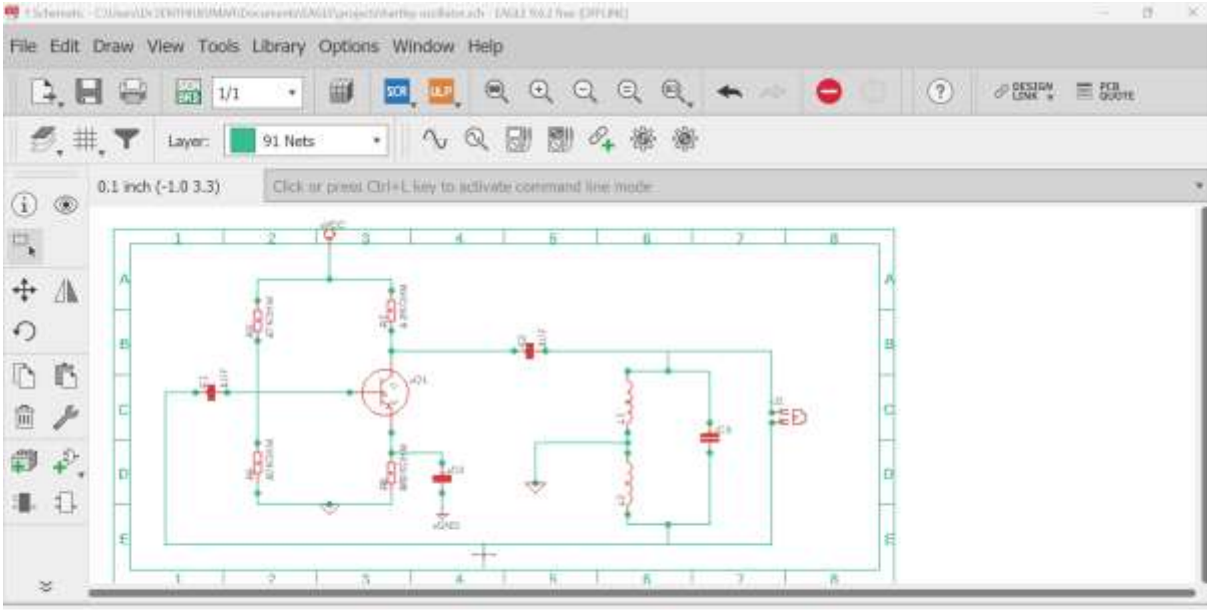


Procedure

Schematic Design

- A new project is created in EAGLE.
- Components such as BF195, resistors, capacitors, and inductors are placed.
- Connections are made using the NET command.
- Electrical Rule Check (ERC) is performed.

PCB SCHEMATIC DIAGRAM



Result

The Hartley oscillator Schematic diagram was successfully designed using PCB software.

EXPERIMENT – 9.2

ROUTE PCB TRACKS WITH MANUFACTURING BOARD FOR HARTLEY OSCILLATOR CIRCUIT USING EAGLE SOFTWARE

Aim

To **route PCB tracks with manufacturing board for** monostable multivibrator schematic using Eagle Software

Apparatus / Software Required

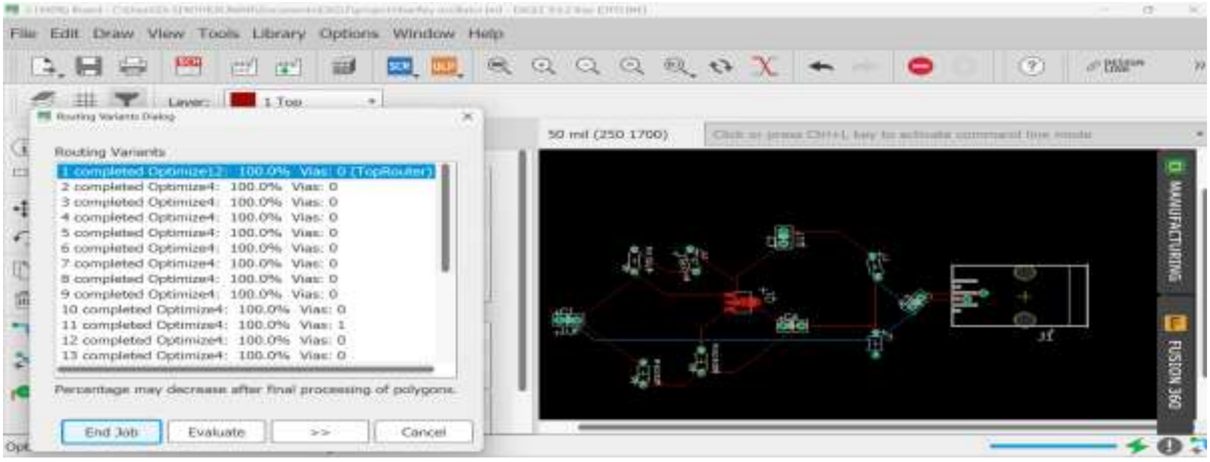
- PCB CAD Software
- PC

Procedure

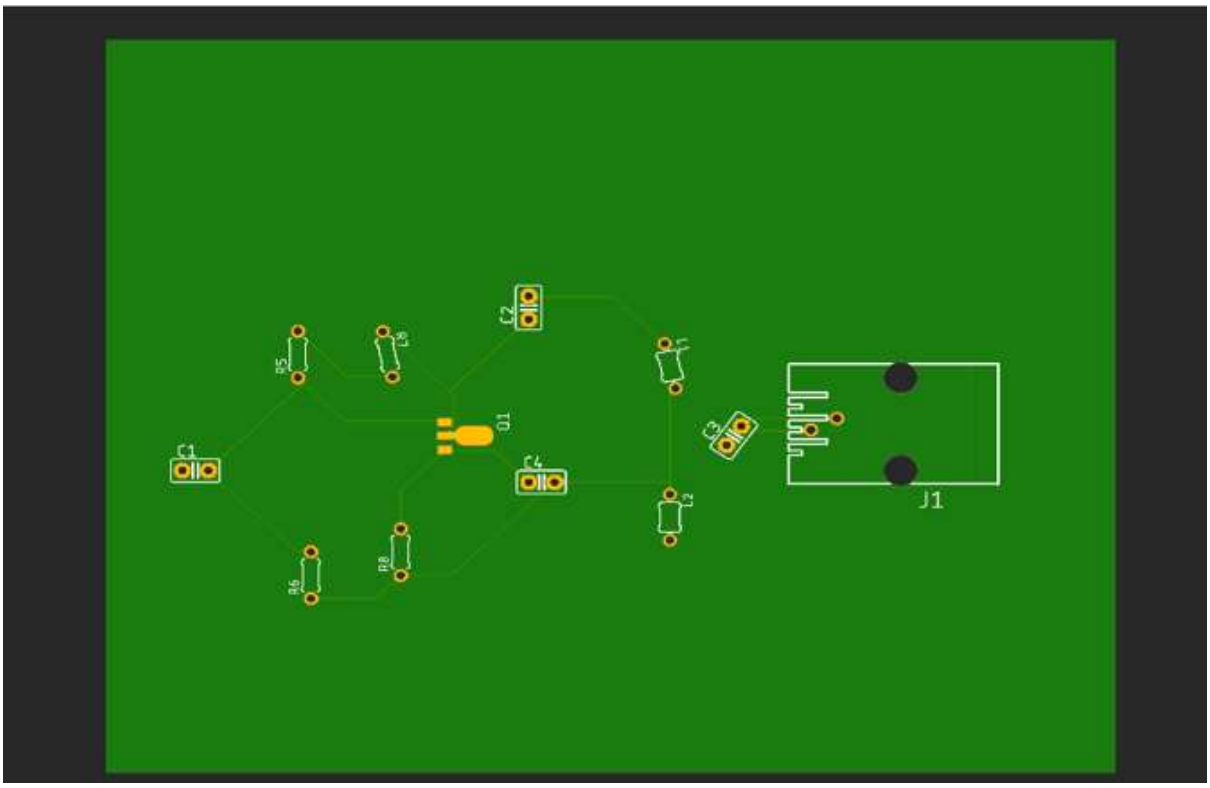
PCB Layout Design

- Schematic is converted to board layout.
- Board outline is defined.
- Components are placed to minimize noise and lead length.
- Traces are routed manually.
- Ground plane is created using POLYGON.

PCB BOARD LAYOUT WITH ROUTING



MANUFACTUREING LAYOUT



RESULT

Thus, **Routing and PCB tracks with manufacturing board for** monostable multivibrator designed successfully.

EXPERIMENT – 9.3

DESIGN OF HARTLEY OSCILLATOR TO GENERATE GERBER FILES USING EAGLE SOFTWARE

Aim

To design a Hartley oscillator circuit and implement its PCB layout using EAGLE software, and to generate Gerber files for fabrication.

Apparatus / Software Required

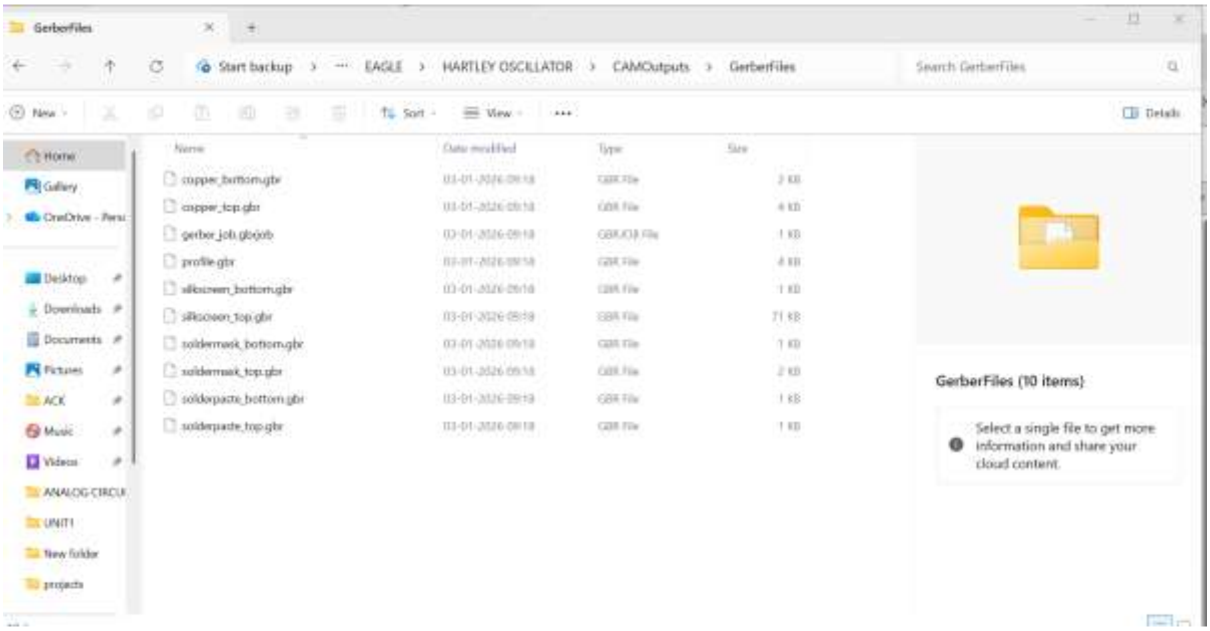
- PCB CAD Software
- Transistor
- Resistors, Capacitors, Inductors
- DC Power Supply (10 V)
- Gerber Viewer

Procedure:

Gerber File Generation

- CAM processor is used.
- Gerber and drill files are generated using RS-274X format.

GERBER FILE DETAILS



Result

The Hartley oscillator circuit was successfully designed using PCB software. The PCB layout was completed and Gerber files were generated for fabrication.

EXPERIMENT – 10

Design of Monostable Multivibrator Using PCB Software

OBJECTIVES

1. Design and Create PCB layout Monostable Multivibrator circuit.
2. Route PCB tracks with Manufacturing board
3. Generate fabrication files.

Expected Learning Outcomes

After completion of this experiment, students will be able to:

1. Understand the operation and applications of a Monostable Multivibrator.
2. Design the schematic of a Monostable Multivibrator using EAGLE CAD software.
3. Calculate pulse width using circuit parameters and timing equations.
4. Develop a PCB layout for the monostable circuit following standard design practices.
5. Perform component placement and PCB routing for reliable circuit operation.
6. Verify the PCB design using Design Rule Check (DRC).
7. Generate Gerber and drill files for PCB manufacturing.
8. Apply PCB design and fabrication concepts for practical implementation of pulse generation circuits.

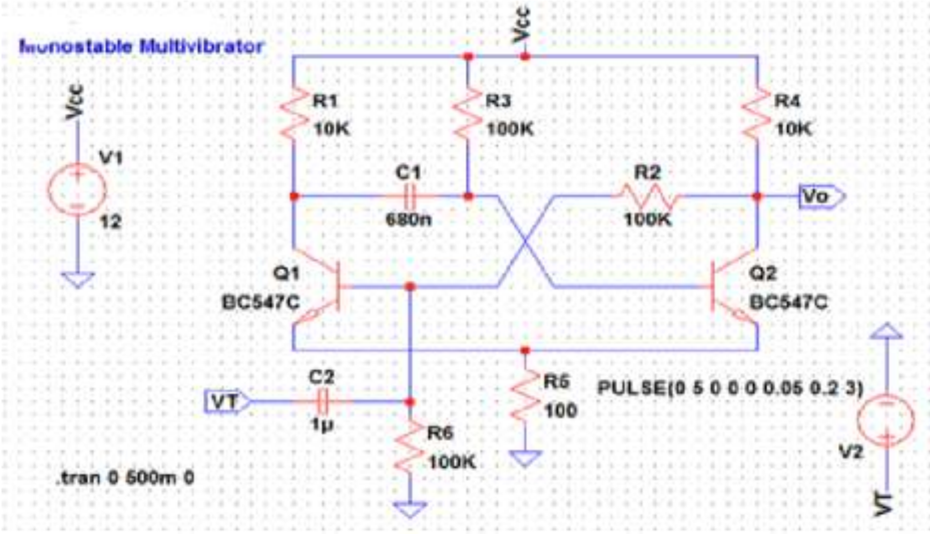
Theory

A monostable multivibrator has **one stable state and one quasi-stable state**.
When a trigger pulse is applied, the circuit switches to the quasi-stable state for a fixed duration and then returns to the stable state automatically.

PulseWidth Equation

T=0.69×R×CT = 0.69 \times R \times CT=0.69×R×C

Circuit Diagram



Experiment 10.1

DESIGN A MONOSTABLE MULTIVIBRATOR CIRCUIT AND IMPLEMENT ITS PCB LAYOUT USING EAGLE SOFTWARE

Aim

To design a **Monostable Multivibrator** circuit and implement its PCB layout using EAGLE software

Apparatus / Software Required

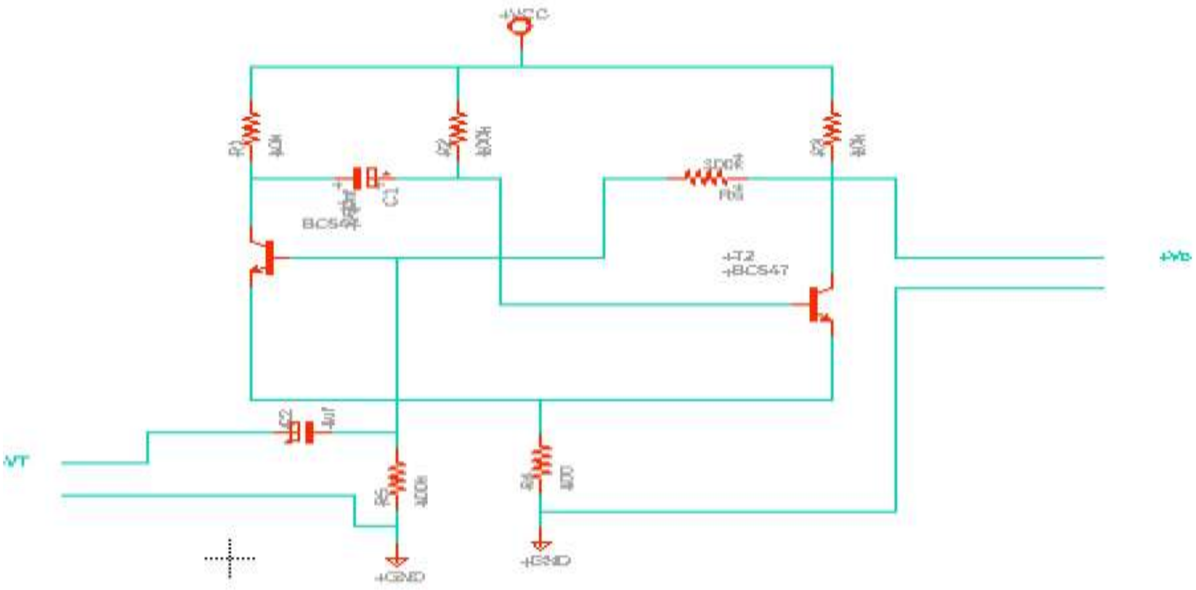
- PCB CAD Software
- PC

Procedure

Schematic Design

- A new project is created in EAGLE.
- Components such as BC547, resistors, capacitors, and inductors are placed.
- Connections are made using the NET command.
- Electrical Rule Check (ERC) is performed.

PCB SCHEMATIC DIAGRAM



Result

The monostable multivibrator Schematic diagram was successfully designed using EAGLE software

EXPERIMENT – 10.2

ROUTE PCB TRACKS WITH MANUFACTURING BOARD FOR MONOSTABLE MULTIVIBRATOR USING EAGLE SOFTWARE

Aim

To **route PCB tracks with manufacturing board for** monostable multivibrator schematic using PCB Software

Apparatus / Software Required

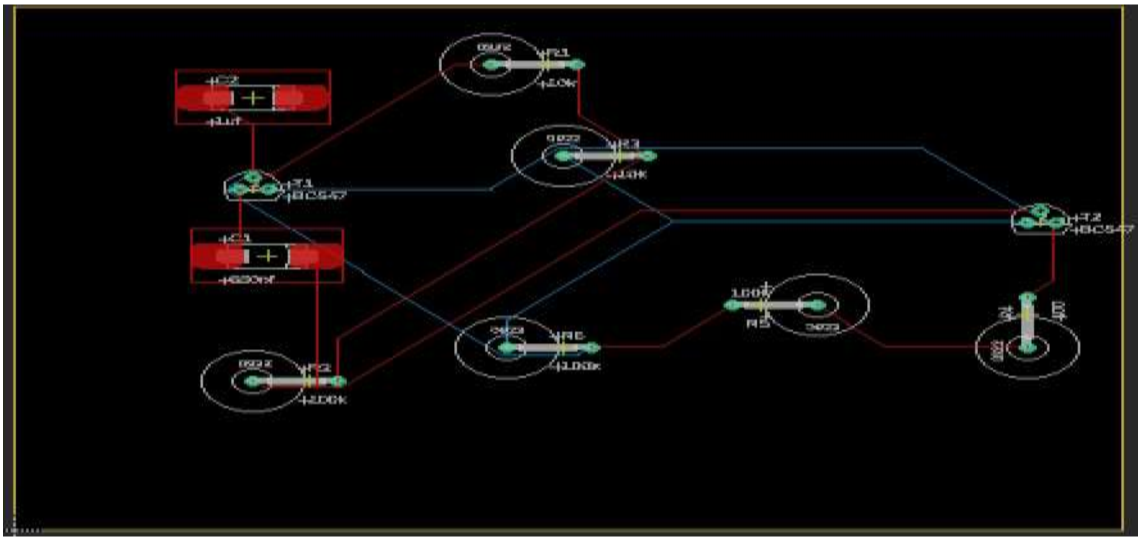
- PCB CAD Software
- PC

Procedure

PCB Layout Design

- Schematic is converted to board layout.
- Board outline is defined.
- Components are placed to minimize noise and lead length.
- Traces are routed manually.
- Ground plane is created using POLYGON.

PCB BOARD LAYOUT WITH ROUTING



MANUFACTUREING LAYOUT

RESULT

Thus, **Routing and PCB tracks with manufacturing board for** monostable multivibrator designed successfully.

EXPERIMENT – 10.3

DESIGN OF MONOSTABLE MULTIVIBRATOR TO GENERATE GERBER FILES USING EAGLE SOFTWARE

Aim

To design a monostable multivibrator circuit and implement its PCB layout using EAGLE software, and to generate Gerber files for fabrication.

Apparatus / Software Required

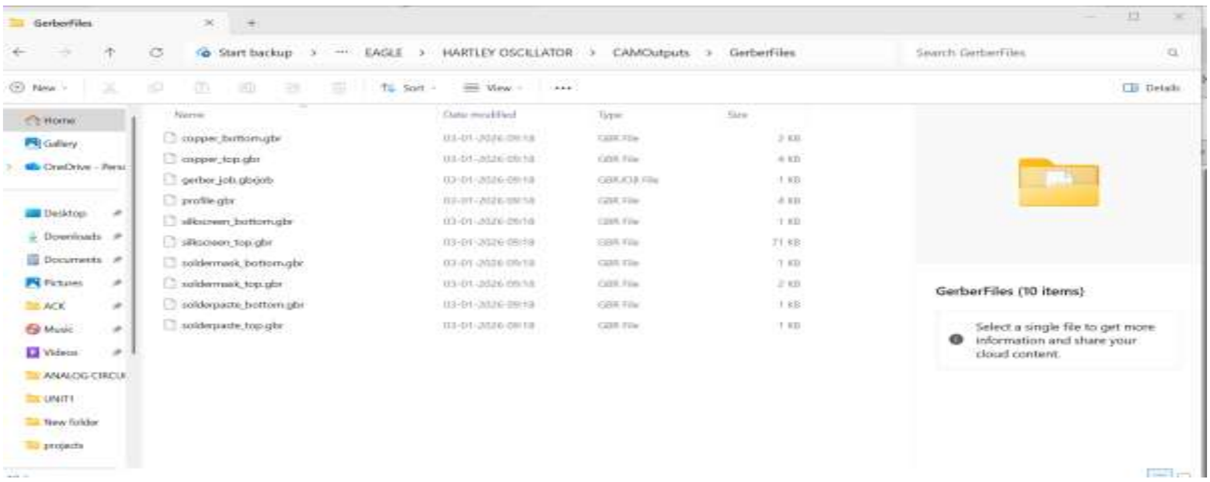
- PCB CAD Software
- PC

Procedure:

Gerber File Generation

- CAM processor is used.
- Gerber and drill files are generated using RS-274X format.

GERBER FILE DETAILS



Result

The Hartley oscillator circuit was successfully designed using PCB software. The PCB layout was completed and Gerber files were generated for fabrication.